

Proxy Advisory Firms and Corporate Shareholder Engagement

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We study how Institutional Shareholder Services (ISS) affect firms' engagement with shareholders. Our analyses exploit a quasi-natural experiment using say-on-pay voting outcomes near a threshold that triggers ISS to review engagement activities. Firms receiving ISS treatment exhibit swift and substantive increases in extensive and intensive margins of engagement, especially when their boards have higher agency conflicts and directors are more likely to lose voting support from ISS. Increases in engagement persist over time, and shareholders appear to value the increased engagement. Collectively, we shed light on an unexplored channel through which ISS positively influences firms' governance and information environments. (*JEL* D72, D82, G34, G38, M12)

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Proxy advisors (PAs) conduct research and provide recommendations to investors that significantly affect voting outcomes (Malenko and Shen 2016). Despite their pivotal role in shaping the information flows that facilitate shareholder voting decisions, the impact of PAs on firm governance and information quality is a topic of debate. Some research suggests that higher-quality PA recommendations can be valuable (Malenko and Malenko 2019; Albuquerque, Carter, and Gallani 2020), while others point to their biased nature, role in standardization of pay plans, reductions in independent information acquisition, and decreases in overall firm value (Larcker, McCall, and Ormazabal 2015; Cabezon 2024; Malenko, Malenko, and Spatt 2024). In line with these concerns, market participants, legislators, and regulators have recently scrutinized the influence of ISS—the largest PA firm—on firms’ governance policies and voting outcomes (Li 2018; SEC 2020; U.S. Congress 2023).

In this paper, we propose an unexplored channel by which ISS likely influences firms’ governance and information environments: *shareholder engagement*. ISS describes engagement as direct discussions between firms and investors on issues such as executive compensation and corporate governance.¹ If firms receive what ISS deems to be a low say-on-pay (SOP) vote, it advises managers to meet with shareholders to better understand their concerns. If managers do not demonstrate a “robust engagement response,” ISS will recommend voting against the firm’s next SOP vote and potentially members of the board.

We posit that ISS can affect corporate governance by encouraging firms to engage with their investors. Such engagement can promote a two-way flow of information, enabling managers and directors to better understand investors’ nuanced views than they would through the coarse one-way signal of voting outcomes. If true, then rather than simply imposing its own standardized guidelines, ISS facilitates richer information exchanges between firms and shareholders and elevates shareholders’ influence on corporate governance and firm policies.

Because public companies are not legally obligated to follow the policies of PAs, it is unclear whether we will find evidence of ISS-induced engagement. However, institutional theory argues that private decentralized organizations like PAs can influence managerial behavior through “economic sanctions,” such as influential voting recommendations (Ingram and Silverman 2002; Delmas and Toffel 2008). Moreover, surveys of corporate directors and institutional investors suggest that PAs play an important role in facilitating firm and investor interactions (McCahery, Sautner, and Starks 2016; Hayne and Vance 2019).

To study shareholder engagement, we develop novel metrics based on firms’ voluntary disclosures in proxy statements, where ISS encourages firms to

¹ See Mehta (2013).

describe their response to a low vote. These metrics reflect the presence (i.e., extensive margins) and intensity (i.e., intensive margins) of discussions using a count of engagement keywords, and whether shareholder discussions reference the previous SOP vote. Additionally, we gauge whether firms summarize shareholder concerns and the board's response in tables, and the proportion of shareholders contacted or spoken with. These measures provide a wide assessment of the breadth and depth of a firm's engagement practices.

We begin our analysis by studying the time trends and determinants of engagement across more than 4,000 firms and over 20,000 SOP votes in the ISS database from 2011 to 2019. In addition to understanding whether PAs, including ISS, play a role in influencing engagement over time, this exploration begins to address the literature gap on the drivers of engagement, which likely stems from the difficulty of observing private conversations. While just 5.5% of firms report shareholder engagement in 2011, it rises to 36.3% by 2019, with an average of 24% during our sample period. This trend coincides with increasing calls for engagement from PAs, shareholders, and regulators.²

In our regressions of the determinants of engagement, we find that firm size, leverage, institutional ownership, analyst coverage, and CEO pay positively correlate with the extensive and intensive margins of engagement. Profitability, market-to-book values, and also certain governance characteristics (classified boards, dual class shares, and compensation committee tenure) are negatively related to engagement. Importantly, we find that SOP voting dissent as well as ISS issuing an "against" recommendation for an SOP proposal both precede higher levels of engagement. Given the effect of ISS voting recommendations, these findings suggest that ISS could strongly influence firms' shareholder engagement policies.

To determine whether ISS's influence on engagement is causal, we exploit ISS's policy mandating a specific response from firms receiving less than 70% SOP voting support, using this threshold as a discontinuity for identification purposes. Should a firm's SOP vote fall below 70%, ISS formally scrutinizes its disclosed shareholder engagement to decide on its voting guidance before the next annual meeting. ISS's policies specify what disclosures are needed for firms to show a robust engagement response to a low vote, which dictates ISS's voting recommendations and its enhanced scrutiny. Failure to show sufficient response could lead ISS to advise against the SOP vote, compensation committee members, or even the entire board.

Our identifying assumption is that firms cannot precisely manipulate the SOP votes around the 70% threshold. This assumption is supported by a smooth distribution at the ISS cutoff, confirmed through graphical analyses

² See Fairfax (2013) for a discussion on the rising demand for engagement. The SEC has also called for greater engagement. For example, then-SEC Chair Mary Schapiro (2010) implored directors to have, "clear conversations with investors with how a company is governed." Similarly, then-SEC Chair Jay Clayton (2018) noted, "Shareholder engagement is a hallmark of our public capital markets."

of SOP plots and a formal robust marginal density test (McCrary 2008). We also examine prior engagement, SOP voting, firm characteristics, CEO pay, and governance before treatment and find sparse statistical differences in these properties between firms just below 70% SOP support (treatment) and those just above (control). These results suggest treatment around this threshold is as good as random, enabling us to attribute subsequent changes in engagement to ISS's influence.

Utilizing linear regression, difference-in-differences (d-i-d), and regression discontinuity (RD) methodologies, we find that ISS treatment leads to a significant increase in the extensive and intensive margins of shareholder engagement in the year after a low SOP vote. The linear regression tests reveal ISS-treated firms are 31% more likely to disclose engagement than control firms. Additionally, treated firms show greater breadth of engagement by contacting and speaking with investors owning a larger number of shares. They also elaborate more on their engagement efforts by offering more comprehensive details in their proxy statements.

These results support the theory that private organizations like ISS can motivate firms to strengthen shareholder engagement using the threat of negative voting recommendations. To further substantiate the causal impact of ISS on engagement, we conduct several cross-sectional analyses. If ISS plays an important role in enhancing engagement and provides what market participants view as a credible threat, we anticipate a noticeably larger engagement response by directors that are at greater risk of economic sanctions. Consistent with this notion, we observe larger engagement increases among treated firms with lower prior voting support for directors as a whole and compensation committee members in particular.

To further test the credibility of ISS's threat, we designate firms as more or less responsive to the SOP vote based on whether they disclose engagement efforts in their subsequent proxy. We find that less responsive treated firms have a greater likelihood of receiving an "against" recommendation by ISS for the entire board, and especially members of the compensation committee. These findings help validate ISS's threat against directors who do not demonstrate a robust response, thereby supporting the mechanism of ISS shaping firms' engagement decisions through economic sanctions, which increases the confidence in our causal inferences.

A natural question arises regarding the sustainability of engagement efforts after treatment. On the one hand, firms could reduce their engagement activities after satisfying ISS's requirements. This finding would be intuitive as elevated ISS oversight ends after demonstrating a robust engagement response, and failure to engage does not result in SEC penalties since SOP votes are nonbinding. On the other hand, treated firms might demonstrate a commitment to engagement to avoid further negative attention from ISS. Moreover, engagement likely involves a one-time cost for establishing engagement protocols and identifying shareholders to engage with. Firms could reap the

benefits of sustained engagement with low marginal costs beyond the first year, potentially leading to greater goodwill among shareholders, improved efficiency of future engagement efforts, and a convergence of beliefs between management and shareholders (Kakhbod et al. 2023). Consistent with the latter thesis, we find that ISS-treated firms show enduring increases in shareholder engagement in the years after, but not before, the low SOP vote.

Next, using an event study around the SOP vote disclosure, we investigate whether ISS-facilitated engagement benefits shareholders. Our findings reveal a positive stock price response when firms both fall just below the 70% treatment threshold and for which ISS had issued an “against” SOP vote recommendation. This stock price response is economically meaningful, with positive abnormal returns between 2% and 3%. These findings suggest that shareholders anticipate that the benefits of strengthened engagement will accrue to investors when the threat of ISS monitoring and economic sanctions is strongest.

Given that markets respond favorably to anticipated engagement, and firms encouraged by ISS to start engaging continue to do so, our final set of tests explore why treated firms did not engage before ISS intervention. In particular, we examine whether the presence of agency conflicts on the board of directors plays a role in engagement policies. We anticipate and find that boards with properties that are typically associated with higher agency conflicts, such as long-tenured compensation committee members and classified boards, increase engagement more after ISS treatment. This finding suggests that ISS can play a substitute governance role for such firms.

Our study faces some limitations. A primary concern relates to our reliance on voluntary disclosures for measuring shareholder engagement, which may not reflect the actual level of engagement. It is comforting, however, that the legal liability of firms for their proxy disclosures to the SEC reduces the likelihood of false or misleading reports.³ Moreover, while not mandated by the SEC, firms have incentives to disclose engagement activities that are perceived by shareholders and PAs as a gauge of board responsiveness. Another important caveat is that while increased engagement facilitated by ISS appears to be beneficial for firms based on the market response, our study is not designed to determine the optimal level of engagement for the firm and its investors.

Our study contributes to several strands of research. Our primary contribution is to the literature on the economic role of PAs (e.g., Alexander et al. 2010; Ertimur, Ferri, and Oesch 2013; Malenko and Shen 2016; Shu 2024), particularly ISS, in shaping corporate engagement policies. We show

³ Directors have a legal obligation to ensure proxy statements are truthful and comprehensive, as mandated by Section 14(a) of the Securities Exchange Act of 1934 and Rule 14a-9. This rule prohibits false or misleading information in proxy statements or omitting material information needed for informed shareholder decisions. See, for example, Kidd (2023).

that ISS's policies promote two-way communication between firms and shareholders after low SOP votes. This greater and sustained engagement likely allows for a more effective exchange of shareholder concerns and managerial strategies beyond simple one-way coarse communication through voting. Our event study results suggest that shareholders value ISS's role in encouraging this dialogue.

Our results on ISS's economic sanctions on directors, and their engagement policies in response, adds to the literature on labor market consequences for directors (e.g., [Fama and Jensen 1983](#); [Srinivasan 2005](#); [Fich and Shivdasani 2007](#); [Del Guercio, Seery, and Woidtke 2008](#)). We show that ISS can incentivize directors to promote engagement policies to preserve their board position and reputation as effective monitors.

Our findings also offer novel insights into ISS's impact on capital markets, challenging the view that PAs simply impose one-size-fits-all pay policies on firms ([Larcker, McCall, and Ormazabal 2015](#); [Jochem, Ormazabal, and Rajamani 2021](#); [Cabezon 2024](#)). In our setting, ISS encourages firms to engage with investors and consider their perspectives based on the unique circumstances of each firm and its investor base. Thus, our results expand on our understanding of how ISS can serve an important governance role ([Malenko and Malenko 2019](#); [Albuquerque, Carter, and Gallani 2020](#)) and complement survey evidence that PAs can improve shareholder engagement activities ([McCahery, Sautner, and Starks 2016](#); [Hayne and Vance 2019](#)).

These results could also inform policy debates, potentially alleviating concerns among regulators and lawmakers about ISS's significant market power and its effects on corporate governance quality ([Financial Economists Roundtable 2022](#); [U.S. Congress 2023](#)). Although our findings cannot answer whether ISS creates or destroys value on net, we show that by encouraging engagement, ISS enables investor preferences to be heard and potentially incorporated in firm policies. In this way, ISS could facilitate better governance decisions for individual firms. Thus, our research could provide valuable insights for contemporaneous discussions on regulating the scope of ISS's voting advisory services (e.g., [Li 2018](#)).

Finally, our paper provides an important contribution to the nascent strand of literature examining engagement between firms and their investors. We introduce a novel disclosure-based approach to provide the first measures of different dimensions of shareholder engagement by firms and document various factors that determine engagement decisions. We complement the findings by [Kakhbod et al. \(2023\)](#), who provide a theory of engagement that, when shareholders and management have different beliefs about corporate policies, the efficacy of shareholder engagement increases when other shareholders also engage. We extend this literature by showing that firms under elevated ISS scrutiny engage more with their investors by strengthening the breadth and depth of engagement.

1. Institutional Background and Hypotheses

1.1 Background information and conceptual framework

PAs play a pivotal role in corporate governance by providing institutional investors with paid guidance on proxy voting matters, based on comprehensive research, analysis, and investor consultations (GAO 2016). They synthesize a large amount of information to make informed recommendations, which can help institutional investors make efficient voting decisions across their diverse portfolios (Ertimur, Ferri, and Oesch 2013; Malenko and Malenko 2019).

The demand for PA services has increased in recent years due to shareholders' needs for informed voting decisions, increased institutional ownership, and regulatory mandates like SOP. The PA advisory market is dominated by ISS and Glass Lewis (GL), which together control more than 90% of the U.S. market (Copland, Larcker, and Tayan 2018; Shu 2024). Research confirms their sizeable influence on voting outcomes (Choi, Fisch, and Kahan 2009; Ertimur, Ferri, and Oesch 2013; Larcker, McCall, and Ormazabal 2015; Malenko and Shen 2016). Specifically, an ISS "against" recommendation for a firm's SOP can decrease shareholder support by 25% (Malenko and Shen 2016), while a negative GL recommendation can lead to a 13% decline (Ertimur, Ferri, and Oesch 2013).⁴

The influence of PAs, however, has drawn regulatory scrutiny and sparked debate over their role in governance, information flows, and firm value (Alexander et al. 2010; Ertimur, Ferri, and Oesch 2013; Malenko and Shen 2016; Larcker, McCall, and Ormazabal 2013, 2015; Shu 2024). While some studies highlight the negative effects, including biased advice, generic policies, crowding out information acquisition, and diminished shareholder wealth (e.g., Iliev and Lowry 2015; Larcker, McCall, and Ormazabal 2015; Li 2018; Malenko and Malenko 2019; Levit and Tsoy 2022; Cabezon 2024; Malenko, Malenko, and Spatt 2024), others suggest high-quality PA recommendations can be beneficial (Malenko and Malenko 2019), particularly during less busy times when more resources are available (Albuquerque, Carter, and Gallani 2020).

Panel A of Figure 1 depicts the assumed traditional information flow under PA voting guidance. The conventional framework for PA influence on firm policies is by selling voting recommendations and advice to institutional investors, helping them make informed voting decisions on proxy ballot items. However, this model suggests a simplistic and potentially incomplete information exchange between shareholders and firms. Additionally, challenges in aggregating information and preferences could curtail the effectiveness of this exchange (Levit, Malenko, and Maug 2024).

We propose shareholder engagement as an alternative mechanism through which PAs can affect the flow of information between firms and investors. Such

⁴ Some argue that PA influence may be due to "robovoting" or the automatic following of recommendations (Malenko and Shen 2016; Shu 2024). Others note that investors may consider PA advice, but do not always follow their recommendations (e.g., Iliev and Lowry 2015; McCahery, Sautner, and Starks 2016).

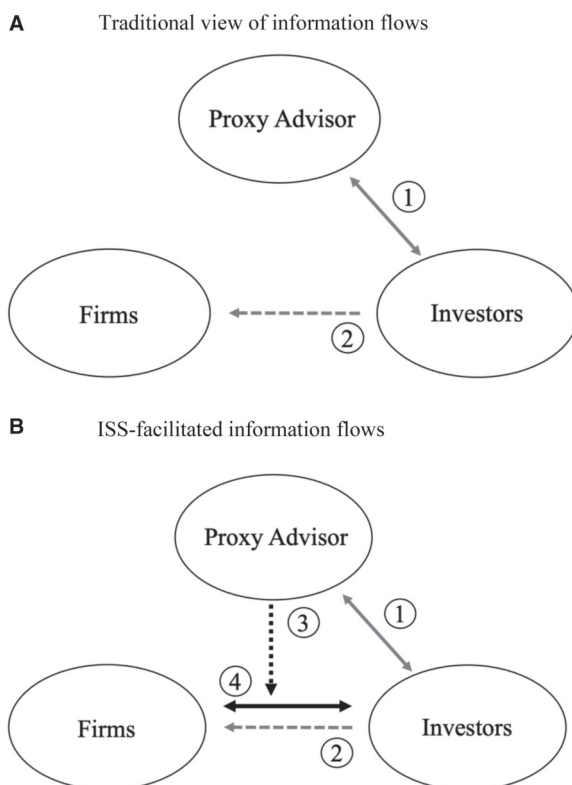


Figure 1

Proxy advisors and information flows

This figure shows how proxy advisory firms help with communication and voting between shareholders and firms. Panel A shows how proxy advisors talk to investors and make recommendations for voting. In step 1, proxy advisors have two-way communication with investors, which helps inform their recommendations. In step 2, shareholders vote on say-on-pay (SOP), but the information content of this vote is coarse as investors can only vote yes, no, or abstain. Panel B shows how, if a firm gets an SOP vote less than 70%, the proxy advisor (ISS) encourages the firm to talk to investors and disclose information about how it addressed shareholder concerns (step 3). We test whether firms respond to pressure by ISS to engage more in direct, two-way communication with investors (step 4).

engagement could promote continuous, two-way communication, enabling firms to better understand shareholder preferences for corporate strategies, beyond simple one-way voting signals. Proactive engagement with shareholders before voting could allow firms to incorporate feedback into decision-making and potentially modify the subsequent voting proposal to sway voting outcomes (e.g., adjust executive compensation for SOP votes).⁵

Our conceptual model suggests that PAs could enhance the integration of detailed shareholder feedback into corporate policies, especially following

⁵ See Gerling et al. (2005) and Malenko (2014) for discussions of prevote communication by boards or committees.

a low SOP vote. This perspective predicts that PAs significantly affect governance and the dynamics of information flows in capital markets, thereby challenging traditional views. Panel B of Figure 1 illustrates our conceptual approach to understanding engagement-driven information exchange under PA guidance.⁶ We explore this theory by examining the engagement policies of ISS, the largest PA firm.

1.2 ISS policy on shareholder engagement

ISS's policies state that, if support for its SOP proposal falls below 70% of votes cast, it will evaluate a firm's responsiveness through engagement. Though ISS does not directly facilitate discussions between firms and shareholders, it requires managers and/or directors to proactively reach out to shareholders and then clearly disclose both quantitative and qualitative aspects about this engagement and subsequent actions taken in response to these concerns in the next annual proxy statement.⁷

ISS also expects firms to disclose the number and share percentage of shareholders they engage with, specific shareholder concerns that led to SOP dissent, concrete actions taken to address these concerns, or explain any reasons for not acting. Additionally, the firm should disclose its plans to prevent similar issues from occurring in the future.

ISS stipulates that if a firm demonstrates a "sufficiently robust response" to shareholder concerns in its proxy statement following a low vote, it will not require future engagement disclosures, provided that future SOP votes achieve at least 70% support. Failure to adequately respond to a low vote will lead ISS to advise against the next SOP proposal and the incumbent compensation committee members. A "moderate" response will result in ISS recommending against only the next SOP proposal, sparing compensation committee members from economic sanctions. Repeated instances of inadequate responses will prompt ISS to recommend voting against the entire board due to systemic issues with board stewardship and oversight.⁸

1.3 Hypotheses development: Level and persistence of shareholder engagement

Although ISS lacks legal authority to mandate firms' engagement practices, our study on its influence over corporate behavior draws from the tenets of

⁶ We thank Nadya Malenko for the insightful depiction and visual representation of our thesis in Figure 1.

⁷ See ISS, U.S. Compensation Policies, December 20, 2018, <https://www.issgovernance.com/file/policy/2019/americas/US-Compensation-Policies-FAQ.pdf>. In conversations with ISS, they note that their engagement policy was established in 2011 after surveying public firms and institutional investors about the level of SOP opposition that should require an explicit response. Figure 3 summarizes the survey results. We provide additional information in Tables IA-1 and IA-2 of the Internet Appendix.

⁸ We verify our interpretation of this policy with ISS and conduct cross-sectional analyses to test the credibility of ISS's policy in our sample (see H2b in Section 2.3).

institutional theory within organizational studies. This theory examines how institutions, including private decentralized entities like ISS, can shape the actions of organizations (Ingram and Clay 2000; Ingram and Silverman 2002; Delmas and Toffel 2008). Despite lacking formal enforcement capabilities, entities like ISS can still affect managerial practices through the potential for economic penalties for failing to comply with its guidelines (Ellickson 1991; Ingram and Silverman 2002; Terlaak 2007).

ISS offers advisory services to institutional investors by informing their votes on shareholder proxy matters in publicly traded companies. While these firms have no legal obligation to follow ISS policies or alter engagement or disclosure practices after receiving less than 70% support in SOP votes, ISS can exert pressure by threatening to recommend against future SOP or director votes for firms that do not demonstrate a robust engagement response in their next proxy filing. Considering ISS's substantial influence on voting outcomes (Malenko and Shen 2016), we expect the prospect of a negative recommendation will motivate firms to increase shareholder engagement and disclose the related information in the subsequent proxy statement following a low SOP vote. This leads to our first hypothesis:

***H1:** Firms with SOP voting support slightly below 70% will follow ISS requirements by increasing shareholder engagement in the following year.*

ISS's guidelines recommend voting against future SOP proposals as well as against the reelection of compensation committee members and the entire board if firms fail to adequately address poor outcomes in SOP votes. Prior research has suggested that labor market incentives can motivate directors to perform well in their roles. Underperforming directors are more likely to face career consequences, including losing current directorships and not obtaining new ones (e.g., Fama and Jensen 1983; Srinivasan 2005; Fich and Shivdasani 2007; Del Guercio, Seery, and Woidtke 2008; Cai, Garner, and Walkling 2009). Given the potential career impact of ISS's economic sanctions, it is plausible that boards at greater risk of such penalties would be more proactive in engaging with shareholders and ensuring these efforts are disclosed in their subsequent proxy statements. To examine these arguments, we propose the following hypotheses.

First, we predict that firms with compensation committee and board members who have already experienced low votes will be more motivated to enhance their engagement efforts due to concerns over ISS's economic repercussions. Formally, we state following hypothesis:

***H2a:** Firms with directors on the compensation committee or the board with low prior SOP voting support will have higher shareholder engagement after ISS intervention.*

Second, should the threat by ISS to impose economic sanctions on unresponsive firms be credible, we expect directors of firms that do not demonstrate a satisfactory engagement response following a low SOP vote will be more likely to receive “against” recommendations from ISS for their reelection. This anticipation of an adverse recommendation on their future candidacy could motivate boards to proactively adhere to ISS’s guidelines. As a result, we predict an increase in negative recommendations for firms that fail to meet ISS’s policy expectations. Thus, we articulate the next hypothesis as follows:

***H2b:** Directors, especially those serving on the compensation committee, of firms subjected to ISS intervention and displaying lower levels of engagement are more likely to face “against” recommendations from ISS for their reelection in the following year.*

If a firm demonstrates robust shareholder engagement following a low vote and achieves a vote above 70% in the next year, ISS will no longer subject it to increased scrutiny. However, firms that are relieved of intense scrutiny may choose to continue their shareholder engagement activities in the longer term. Prior to receiving a low SOP vote, firms may have been reluctant to engage with shareholders, believing the costs of engagement to exceed the benefits. These costs could include the initial fixed setup expenses for creating an engagement infrastructure, such as defining engagement protocols for which firm personnel will engage, identifying the shareholders to interact with, developing the necessary technology for communication, and organizing legal guidelines.

After the initial year of ISS-induced engagement, and once the engagement infrastructure is in place, the marginal benefits, such as improved information flow quality, and the development of goodwill and supportive shareholder relationships (Kakhbod et al. 2023), may outweigh the marginal costs of maintaining this engagement infrastructure. Further, the initial interactions might more closely align the perspectives of management and shareholders (Crawford and Sobel 1982), improving the effectiveness of future engagements, which are more likely to be initiated by management on a voluntary basis. Other factors for sustaining engagement could be that firms might also aim to maintain their shareholder support above the 70% threshold to avoid negative scrutiny from ISS or shareholders in the future. Additionally, directors may want to maintain their reputation as effective monitors and could be motivated to show their commitment to engagement and transparency to bolster their reelection prospects. With these factors in mind, we propose the following hypothesis:

***H3:** ISS-induced increases in shareholder engagement following SOP voting support just below 70% will not be short-lived.*

If the evidence supports H3, it would be consistent with findings from surveys conducted by Hayne and Vance (2019). Their research finds that both directors and compensation committees believe that PAs can enhance firm communication by “providing an impetus for firms to engage with institutional investors.”

Several forces, however, could obscure our ability to find evidence supporting H1, H2, or H3. First, since advisory votes on executive compensation are nonbinding, firms do not face SEC penalties for low approval rates of their executive pay policies. Second, firms with just over 70% approval may react in a similar manner to those with slightly less than 70% approval, as both are within the lowest quintile for support of their compensation practices.⁹ Third, firms slightly above the 70% threshold might anticipate falling below the threshold in future SOP votes and could thus increase engagement with shareholders to avoid negative attention from ISS. Fourth, receiving slightly less than 70% approval still has the majority of shareholders backing its executive pay practices, potentially leading firms to not increase their engagement efforts unless SOP support falls below 50%. Lastly, the perceived costs of increasing engagement might deter managers and/or directors from increasing their engagement levels.

It is also possible that evidence might support H1 and H2, but not H3. This scenario could take place if firms increase their interactions with shareholders following a poor SOP vote outcome merely to align with ISS guidelines. After meeting ISS’s guidelines for a robust engagement response, the firm might consider the compensation issue resolved and return to its previous level of shareholder engagement.

1.4 Hypotheses development: Shareholder reactions to ISS treatment

We explore whether ISS-induced engagement is beneficial through an event study centered on the announcement of SOP results. Forced intervention by ISS may result in wasteful use of managerial time and resources, along with potentially value-diminishing policy adjustments (Larcker, McCall, and Ormazabal 2015). Such a scenario could result in negative announcement returns for treated firms. It is also possible that firms meet with shareholders, but investors expect any changes to compensation or governance policies to be immaterial. In this scenario, we might not observe an announcement return that is significantly different from zero. Another possible outcome is that shareholders expect ISS’s policies will foster greater communication between managers and shareholders that generate value for the firm. In this case, we might observe positive and significant returns for treated firms. Given the competing forces, we state the following hypothesis in the null form:

⁹ Over the 2011–2019 period, firms received an average of 91% support for SOP and fewer than 2% of these votes failed to receive a majority support. Therefore, even without ISS’s policy, firms on both sides of the 70% SOP support threshold may try to increase shareholder engagement and show that they are responsive to investor concerns.

H4a: *Treated firms will not experience significant returns around the SOP vote announcement.*

It is possible that a significant shareholder reaction might only occur when the looming threat of ISS enforcement is the strongest. In this scenario, we could observe positive stock returns when a firm is both treated *and* previously received an “against” SOP recommendation from ISS, indicating a heightened risk of economic sanctions against board members. Thus, as an extension, we test this hypothesis:

H4b: *Treated firms will experience positive returns when they have already received “against” recommendations from ISS.*

Evidence supporting H1 through H4 would strengthen our argument that ISS plays a crucial role in capital markets by promoting better shareholder engagement, enabling ongoing and bidirectional information exchange between firms and investors. Such findings could also attenuate concerns that ISS’s policies lead to a one-size-fits-all approach to governance and compensation. Instead, evidence in support of H1 to H4 would be consistent with the notion that ISS motivates managers and directors to interact with shareholders and address diverse issues unique to each firm.

2. Sample and Engagement Measures

2.1 Sample selection

We construct our sample by downloading all SOP voting outcomes from the ISS Voting Analytics database over calendar years 2011 to 2019. From this sample, 20,610 firm-years for 4,075 unique firms are matched with proxy statement information on the SEC’s website. We use this sample for two types of analysis. For the first analysis, aimed at identifying engagement determinants, we include all 20,610 firm years in which a firm discloses SOP voting results.

For the second analysis, we focus on a subsample of 426 firms that, for the first time during our sample period, disclose an SOP vote within the narrow $\pm 2.5\%$ range around the 70% support level.¹⁰ We define *SOP voting support* following the ISS definition of votes for the SOP proposal divided by the votes for and against SOP. Abstention votes are not included in the ISS definition; thus, we do not include them in our definition of voting support. The selection criteria for the second analysis allows for clean testing of our hypotheses. To ensure the completeness of our sample, we cross-verify our data set against a file provided directly by ISS that lists all firms with SOP votes under 70%.

We merge these data with the Center for Research in Security Prices (CRSP) and Compustat databases and exclude observations with missing stock price or

¹⁰ In our RD analyses, we verify our results are robust to wider bandwidths around the 70% threshold.

accounting information used as controls. Our sample for this second analysis contains 209 firms with votes just below the 70% threshold (treated firms) and 217 firms just above (control firms). The sample is approximately equally distributed across our sample period. We define *ISS treatment* as an indicator variable that equals one for firms with SOP voting support between 67.50% and 69.99%, and zero for those with support between 70.00% and 72.50%. Appendix A provides a timeline of ISS treatment and subsequent changes in shareholder engagement, pay disclosure, and the event study, all of which we will define below.

2.2 Measures of shareholder engagement

2.2.1 Extensive and intensive margins of engagement Our shareholder engagement analysis relies primarily on textual analysis of proxy statements, where ISS encourages firms to voluntarily disclose engagement activities so they can monitor the response. We provide examples of engagement disclosure in Appendix B. To construct these measures, we develop a Python script to extract text from the proxy statement. Our script then flags engagement keywords (*engage, feedback, conversation*) and their variants within a 100-character range of shareholder keywords (*shareholder, stockholder*) and their variants. For the subsample of firms in the $\pm 2.5\%$ range around the 70% SOP voting support, we hand check the proxy statements for accuracy, but our inferences are unchanged if we rely solely on the script.¹¹

From this keyword identification process, we generate two variables to capture the extensive and intensive margins of shareholder engagement during our 9-year sample period. For our extensive measure, *engagement indicator* equals one if the firm mentions any engagement keyword near a shareholder keyword in the proxy statement, and else zero. This measure captures the presence of engagement following the prior shareholder meeting. For our intensive measure, *engagement count* is the number of engagement keywords near shareholder keywords within the proxy statement. We assume that more engagement discussions in the proxy reflects greater engagement intensity.

Table 1 and Figure 2 display time trends in shareholder engagement for the full sample. We find that at least one mention of shareholder engagement was disclosed for 5.7% of firms in 2011, rising to 16.0% in 2012, and monotonically increasing to 37.0% in 2019. The intensity of engagement also grows over time, starting at an average of 0.18 engagement words per firm in 2011 and increasing to 1.26 by 2019. During the sample period, 23.4% of firms in the ISS database disclose the presence of shareholder engagement, averaging 0.66 engagement words. When engagement occurs, firms provide an average of 2.62 engagement words in their annual proxy during the sample period (untabulated).

¹¹ Firms sometimes provide engagement measures in images that cannot be scraped via a Python code. To ensure a greater level of accuracy, we hand-check engagement discussions for the sample of 426 firms near the 70% threshold. Results using only the Python scrape are reported in Table IA-3 of the Internet Appendix.

Table 1
Shareholder engagement measures over time

	(1)	(2)	(3)	(4)	(5)	(6)
	Engagement indicator	Engagement count	Engagement references SOP	Engagement table %	Contacted % (nonzero)	Spoken with % (nonzero)
2011	0.057	0.180	0.010	0.000	—	—
2012	0.160	0.357	0.030	0.000	0.700	0.250
2013	0.188	0.433	0.100	0.000	0.557	0.495
2014	0.220	0.537	0.100	0.010	0.407	0.498
2015	0.269	0.705	0.100	0.000	0.552	0.565
2016	0.294	0.862	0.120	0.020	0.517	0.402
2017	0.316	0.928	0.100	0.030	0.510	0.473
2018	0.353	1.156	0.110	0.030	0.543	0.486
2019	0.370	1.257	0.220	0.040	0.507	0.336
2011–2019	0.234	0.657	0.099	0.014	0.521	0.446

This table presents measures of shareholder engagement reported in annual proxy statements between 2011 to 2019. *Engagement indicator* equals one if the proxy statement mentions any of the engagement keywords engage*, feedback*, and conversation* within 100 characters on either side of the shareholder keywords shareholder* or stockholder*, and otherwise zero. *Engagement count* represents the number of instances in which engagement keywords appear near shareholder keywords in the proxy statement. *Engagement references SOP* equals one if the shareholder engagement discussion in the proxy statement references the SOP vote from the prior year, and otherwise zero. *Engagement table* equals one if the firm provides a tabular summary of investor concerns and the board response following engagement in the proxy statement, and otherwise zero. *Contacted* is the percentage of shares outstanding held by investors that were contacted during engagement, from the proxy statement. *Spoken with* is the percentage of shares held by investors spoken during engagement, from the proxy statement. *Engagement indicator* and *engagement count* were collected via a python scrape of proxy statements for firms in the ISS Voting Analytics database over 2011 to 2019. *Engagement references SOP*, *engagement table*, *contacted*, and *spoken with* are hand-collected from 900 randomly selected proxy statements uniformly distributed across the sample period.

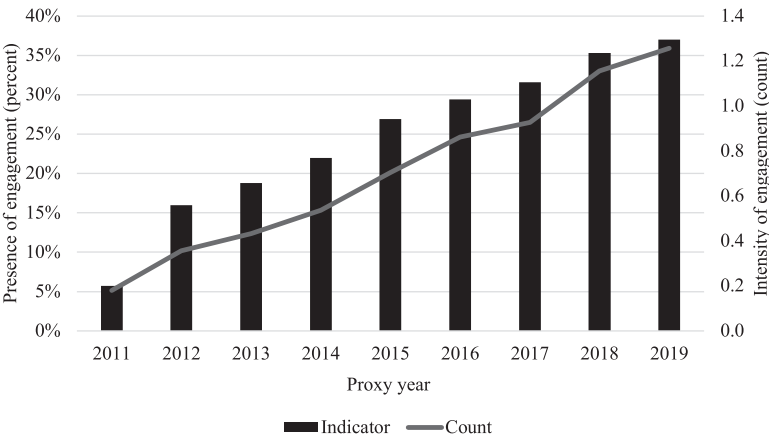


Figure 2
Time trends in shareholder engagement

This figure graphs our primary measures of shareholder engagement for the full universe of firms in the ISS Voting Analytics database with say-on-pay votes over the proxy years 2011 to 2019. Bars represent the yearly sample average for the *Engagement indicator* (left y-axis). Line depicts the yearly average of the *Engagement count* (right y-axis). We define variables in Appendix C.

2.2.2 Depth and breadth of engagement To capture engagement depth, we hand collect two points of information from the proxy statement in the year after the low SOP vote (year $t+1$). Some firms note that their engagement efforts are a direct response to a low SOP vote in the prior year. Thus, we create the variable, *engagement references SOP*, which equals one if the proxy statement links the firm's engagement efforts to the prior SOP vote, and else zero. As shown in Appendix B, firms sometimes provide tables that follow the format of "what we heard" and "our response" to investor concerns that led to voting dissent. A table highlighting shareholder concerns and the firm's responses could demonstrate that the firm takes engagement seriously, potentially boost investor confidence and reduce information processing costs.¹² Thus, we create a variable, *engagement table*, which equals one if the firm includes a summary table of its engagement activity, and otherwise 0.

Some firms highlight the extent of shareholder engagement based on the holdings of investors they contact or speak with. Thus, to assess engagement breadth, we measure (1) the percentage of shares owned by investors contacted after a low SOP vote (*shareholders contacted*) and (2) the percentage of shares held by investors who agreed to speak with the firm (*shareholders spoken with*). Both variables offer precise insight into the firm's outreach efforts and enhance the credibility of our textual-based engagement proxies. Not all firms disclose these data. For those that do not and have an *engagement indicator* of zero, we set both *shareholders contacted* and *shareholders spoken with* to zero. If a firm has an *engagement indicator* of one but lacks disclosure of *shareholders contacted* or *shareholders spoken with*, we classify these values as missing. Table IA-4 in the Internet Appendix reveals that our engagement measures are highly, but not perfectly, correlated. Thus, measuring various aspects of shareholder engagement helps us capture firms' actions more comprehensively.

Since the measures of the depth and breadth of engagement must be hand collected, we randomly select 900 proxy statements uniformly across the sample period. Table 1 shows an increase in firms referencing the prior year's SOP vote, increasing from 1% in 2011 to 22% in 2019. However, there was little change in the percentage of *shareholders contacted* or *shareholders spoken with* during this period. Engaging firms contact and converse with holders of approximately 52% and 45% of shares outstanding, respectively, when these metrics are reported. Engagement tables are rare in the overall sample, with only 1.4% presenting these details. However, as shown in Table 5 below, 19% of the firms just below the 70% SOP threshold provide engagement tables, indicating this practice is more common after lower SOP voting support.

¹² Dong et al. (2016) find the SEC's adoption of XBRL, which organizes financial statement information, lowers the cost of acquiring and processing information for investors. Anecdotally, we observe that shareholder feedback and firm actions were clearer when provided in summary tables rather than in narrative text within the proxy statement.

3. Determinants of Engagement

3.1 Potential determinants

This section explores factors that might influence shareholder engagement using the full sample of 20,610 firm years. Given that our study introduces novel engagement measures, the goal of this exercise is to lay the groundwork for understanding the dynamics of shareholder engagement and its association with firm attributes. Although our empirical framework cannot exogenously identify all determinants, this analysis offers valuable insights that can help validate our engagement measures and bolster our analysis of proxy advisors' impact on shareholder engagement. Additionally, it provides novel descriptive evidence on a trend that has grown significantly in capital markets, as highlighted in Table 1 and Figure 2.

In choosing the determinants to analyze, we first look to factors identified in prior studies of SOP voting (e.g., Malenko and Shen 2016). For example, we examine how engagement relates to the firm's SOP voting outcomes and recommendations from PAs. Using ISS Voting Analytics data, we generate the variable, *ISS "against" SOP*, which equals one if ISS recommends voting against SOP, and else zero. We follow Shu (2024) in obtaining GL data for a subsample of 20% of firm years using a Freedom of Information Act request from the North Carolina Retirement System. We generate a variable, *GL "against" SOP*, which equals one if GL recommends voting against SOP and else zero. We anticipate that firms with negative PA recommendations and higher SOP voting dissent will engage more with investors in the subsequent year.

We expect that firms with more institutional ownership and analyst following will have stronger incentives to engage with shareholders due to higher demand for firm information (Boone and White 2015). We use IBES Academic data and define $\ln(analysts)$ as the log-transformed value of one plus the number of analysts providing earnings per share forecasts during the four quarters prior to the annual meeting. Data on institutional ownership are from SEC 13-F filings in the Thomson Reuters Institutional Holdings database. We define total *institutional ownership (IO)* as the total shares held by these institutions divided by shares outstanding from CRSP. The type of institutional investors (i.e., passive versus active) might affect engagement (Kakhbod et al. 2023). Therefore, we separate the proportion of IO into different types of institutional investors (*dedicated*, *transient*, or *quasi-indexer*) based on portfolio turnover, diversification, and investment horizons as described in Bushee (1998).

Our determinants analysis also includes other factors that might influence shareholder engagement tendencies. We expect that firm characteristics, such as *size*, *market-to-book*, *return on assets*, *leverage*, *sales growth*, *abnormal returns*, *stock return volatility*, and *operating loss*, might affect a firm's propensity to engage with shareholders about components of executive

compensation. We merge these measures using data from CRSP and Compustat and define all variables in Appendix C.

The SEC requires most firms to provide a section in their annual proxy statement called the Compensation Disclosure & Analysis (CD&A), that details information about executive pay, including the goals, design, and reasoning behind the pay program, and the use of peer benchmarking. The SEC allows small firms that meet the definition of a Smaller Reporting Company (SRC) to forego detailed disclosures about executive pay, but they must still hold an SOP vote. We label these firms as *CD&A exempt* using an indicator variable if it had a public float less than \$75 million before September 10, 2018, or less than \$250 million after that date (SEC 2018). We manually check annual SEC Form 10-Ks and identify 33 sample firms as SRCs. The expected impact of not providing a CD&A on engagement is unclear. While the absence of a CD&A could increase engagement efforts to clarify compensation policies—as Balsam et al. (2016) link CD&A prominence to SOP voting results—it might also result in reduced engagement due to the exemption from CD&A requirements.

Firms with higher CEO pay might engage more with shareholders to explain the components of executive pay. We partition pay levels into *CEO equity pay* and *CEO nonequity pay* because shareholders might be more concerned about how CEOs are compensated, rather than the overall level of pay.

Firms with stronger board oversight and governance of executive pay may engage more with shareholders. To proxy for oversight and governance, we follow studies (e.g., Knyazeva, Knyazeva, and Masulis 2013) in measuring *board independence*, *classified boards*, and *dual class shares* for firms in the ISS Governance database. Since U.S. exchanges require almost all firms to have a fully independent compensation committee, we measure pay governance using the average *compensation committee tenure*, which prior work links to higher CEO pay (Vafeas 2003). Finally, we assess whether *CEO turnover*, which is known to influence pay changes (Denis, Jochem, and Rajamani 2020), alters a firm's engagement policies in the prior year.

3.2 Determinants results

We examine the determinants of shareholder engagement via two types of analyses. The first is a univariate analysis of the relation between the extensive margins of engagement and each individual firm or board characteristic. The purpose of this exercise is to see how each factor relates to engagement in isolation. In our second analysis, we examine both the extensive and intensive margins of engagement in a multivariate setting to identify the potential overall drivers of engagement.

3.2.1 Univariate analysis of engagement We examine correlations between the *engagement indicator* and characteristics of 20,610 firm years during 2011 to 2019. We sort the sample into quintiles based on yearly firm-level

Table 2
Determinants of shareholder engagement

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lowest	Q2	Q3	Q4	Highest	Q5-Q1	p-value	Coverage
<i>SOP voting dissent</i>	0.215	0.276	0.299	0.327	0.485	0.270***	.000	1.000
<i>ISS "against" SOP</i>	0.293				0.531	0.238***	.000	1.000
<i>GL "against" SOP</i>	0.431				0.642	0.211***	.000	0.199
<i>Size</i>	0.039	0.156	0.250	0.324	0.456	0.417***	.000	1.000
<i>Market-to-book</i>	0.244	0.348	0.357	0.343	0.304	0.060***	.000	0.989
<i>Return-on-assets</i>	0.200	0.296	0.295	0.371	0.343	0.144***	.000	0.994
<i>Leverage</i>	0.234	0.247	0.288	0.383	0.390	0.156***	.000	0.991
<i>Sales growth</i>	0.309	0.318	0.326	0.346	0.314	0.005	.661	0.979
<i>Abnormal returns</i>	0.303	0.336	0.341	0.333	0.327	0.024**	.033	0.965
<i>Stock return volatility</i>	0.378	0.370	0.316	0.299	0.247	-0.131***	.000	0.963
<i>Operating loss</i>	0.285				0.352	0.067***	.000	0.996
<i>CD&A exempt</i>	0.360				0.084	-0.276***	.000	0.994
<i>Analysts</i>	0.228	0.283	0.331	0.405	0.505	0.277***	.000	0.757
<i>Inst. ownership (IO)</i>	0.095	0.115	0.274	0.393	0.418	0.323***	.000	0.860
<i>Quasi-indexer IO</i>	0.108	0.129	0.273	0.390	0.414	0.306***	.000	0.848
<i>Transient IO</i>	0.094	0.195	0.355	0.369	0.395	0.301***	.000	0.848
<i>Dedicated IO</i>	0.221	0.146	0.315	0.463	0.396	0.174***	.000	0.848
<i>CEO turnover</i>	0.315				0.439	0.124***	.000	1.000
<i>CEO equity pay</i>	0.283	0.284	0.381	0.506	0.485	0.201***	.000	0.680
<i>CEO nonequity pay</i>	0.284	0.331	0.396	0.516	0.422	0.138***	.000	0.680
<i>Board independence</i>	0.377	0.380	0.417	0.435	0.488	0.111***	.000	0.497
<i>CC tenure</i>	0.440	0.444	0.419	0.423	0.349	-0.091***	.000	0.496
<i>Classified board</i>	0.446				0.344	-0.102***	.000	0.536
<i>Dual class shares</i>	0.415				0.270	-0.146***	.000	0.536

This table examines the univariate determinants of shareholder engagement. We sort the sample into quintiles based on yearly firm-level characteristics in fiscal year t for 20,610 firm years during the sample period of say-on-pay (SOP) votes over 2011 to 2019. We then compute the percentage of firms where the *Engagement indicator* equals one within each quintile in year $t+1$. We compute the difference in the highest and lowest quintile (Q5 minus Q1) for each variable. We report the associated p -value and percentage of the sample with coverage (i.e., nonmissing values) for each firm characteristic. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed t -tests).

characteristics in fiscal year t . We then compute the proportion of firms engaging with shareholders within each quintile in year $t+1$, which we present in columns 1 to 5 of Table 2. For each variable, we compare the highest and lowest quintile (Q5 minus Q1) in column 6 and test whether this difference is statistically significant using two-tailed t -tests, with the associated p -value reported in column 7. Since we do not have full data on all measures for all sample years, we report the percent coverage of each variable in column 8.

The results show that engagement varies substantially within the quintiles of each firm characteristic. For example, 21.5% of firms in the lowest quintile of SOP voting dissent engaged with investors while 48.5% of those in the highest quintile engaged with shareholders. The difference of 27 percentage points is significantly different from zero at the 1% level.

Firms with recommendations to vote against SOP by ISS or GL have a significantly higher propensity to engage. For these firms, the extent of engagement is 23.8- and 21.1-percentage-points higher, respectively.

These tests provide initial evidence that PA recommendations could influence engagement.¹³

The rest of panel A shows that engagement varies across firm characteristics. Firms that are larger (*size*), more profitable (*return-on-assets*), have less *stock return volatility*, more analyst coverage, and greater *leverage* tend to engage more and this relation is mostly monotonic across the quintiles. Firms with higher growth opportunities (*market-to-book*) and *abnormal returns* engage more, but the relation is not increasing across all quintiles. Firms with *CEO turnover*, an *operating loss*, and those that are not *CD&A exempt* have higher engagement, however, there is no difference among the highest and lowest quintiles of *sales growth*.

Firms with more overall *institutional ownership* tend to have a higher propensity to engage. This relation appears to be monotonic across the quintiles. Because we observe similar variation in engagement among the different types of institutional ownership (i.e., dedicated, transient, and quasi-indexer), we include overall institutional ownership as a control in our multivariate analysis.

Firms with higher CEO pay (equity and nonequity) also tend to engage more with investors. The difference in the engagement propensity between the highest and lowest quintile for CEO equity pay and CEO nonequity pay is 20.1 and 13.8 percentage points, respectively. Engagement is lower for firms with a lower proportion of independent directors, dual class shares, classified boards, and where members of the compensation committee have longer tenure.

3.2.2 Multivariate analysis of engagement Next, we explore the determinants of shareholder engagement using multivariate regressions. For these tests we estimate the following equations:

$$y_{it+1} = \alpha + \beta \text{ SOP voting dissent}_{it} + X_{it} + \varphi_j + \mu_t + \varepsilon_{it}, \quad (1a)$$

$$y_{it+1} = \alpha + \beta \text{ ISS "against" SOP}_{it} + X_{it} + \varphi_j + \mu_t + \varepsilon_{it}, \quad (1b)$$

where the dependent variable y_{it+1} includes the two primary measures of engagement (*engagement indicator* and *engagement count*). The coefficients of interest are β , which captures how engagement relates to SOP voting (Equation (1a)) and the SOP voting recommendation of ISS (Equation (1b)) in the prior year. To avoid multicollinearity, we do not include both *SOP voting dissent* and *ISS "against" SOP* in the same regression since ISS "against" recommendations are strongly correlated with the SOP voting dissent (69%) and are causally linked to SOP voting outcomes (Malenko and Shen 2016).

¹³ The remainder of our paper focuses on the causal effect of ISS on shareholder engagement. We do not further study the influence of GL on shareholder engagement because of data limitations on GL recommendations, and because GL has evolving SOP engagement thresholds, less market power than ISS, and weaker threats for enforcing the failure to meet GL's engagement expectations. We discuss GL policies in Section 1.1 of the Internet Appendix.

Both equations include fixed effects (FEs) for the firm's industry (ϕ_j) using one-digit Standard Industrial Classification (SIC) codes, which controls for time-invariant unobserved heterogeneity across industries, and fiscal year FEs (μ_t) to control for time trends that affect all firms. We also estimate Equations (1a) and (1b) by replacing industry FEs with firm FEs. Standard errors are double clustered at the year and industry levels for regressions with industry FEs, and at the year and firm levels for those with firm FEs.

X_{it} is a vector of firm-level covariates. These include the natural logarithm of *CEO equity* and *nonequity pay*, *board independence*, *compensation committee tenure*, *classified board*, *size*, *market-to-book*, *return-on-assets*, *leverage*, *sales growth*, *abnormal returns*, *stock return volatility*, *operating loss*, *CD&A exempt*, *ln(analysts)*, *institutional ownership*, and *CEO turnover*. We measure accounting variables in the same fiscal year as reported CEO pay.

Table 3 reports the results. Columns 1 through 4 show the tests of the *engagement indicator*, which is estimated using ordinary least squares (OLS) regressions. We use a linear probability model rather than a probit or logit to avoid bias from FEs (Greene 2004). Columns 5 through 8 report the determinants of *engagement count*, which is estimated using a Poisson regression (Cohn, Liu, and Wardlaw 2022).

The variables of interest, *SOP voting dissent* and *ISS "against" SOP*, both have a strong and positive correlation with the *engagement indicator*. Column 1 shows that after controlling for other firm characteristics, the coefficient for *SOP voting dissent* is 0.955, meaning that an increase of 10 percentage points against SOP is associated with a 9.6-percentage-point higher probability of firms engaging with shareholders. Similarly, the coefficient of 0.316 on *ISS "against" SOP* in column 2 indicates that a firm is 32-percentage-points more likely to engage with shareholders in the following year if ISS recommends against its SOP vote. Both coefficients are statistically significant at the 1% level and are economically meaningful.

We also find significant relations between the variables of interest and *engagement count*. In column 5, the coefficient for *SOP voting dissent* is 2.776, meaning that an increase of 10 percentage points in *SOP voting dissent* is associated with an increase in *engagement count* of 1.51 instances.¹⁴ In column 6, the coefficient of 1.040 on *ISS "against" SOP* shows this recommendation is associated with an increase in *engagement count* of 1.81 instances. These relations are similar when industry FEs are replaced with firm FEs in columns 3 and 4 for the *engagement indicator* and columns 7 and 8 for *engagement count*.¹⁵

¹⁴ A 10% change in engagement count is calculated as $10\% \times (e^{2.776} - 1)$.

¹⁵ In Table IA-5 of the Internet Appendix, we report a similar relation between *GL "against" SOP* and shareholder engagement. We do not include GL recommendations in Equation (1b) as do not have coverage for 80% of sample firm years. The correlation between *GL "against" SOP* and *ISS "against" SOP* is 34%, which could bias results due to multicollinearity. However, we find similar results after including both *ISS "against" SOP* and *GL "against" SOP* in Equation (1b) and replacing *ISS "against" SOP* with *GL "against" SOP* in Equation (1b).

Table 3
Regression analysis of shareholder engagement determinants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Engagement indicator				Engagement count			
<i>SOP voting dissent</i>	0.955*** (14.73)		0.541*** (11.17)		2.776*** (35.72)		1.740*** (13.10)	
<i>ISS “against” SOP</i>		0.316*** (10.97)		0.174*** (8.92)		1.040*** (27.83)		0.583*** (11.64)
<i>Size</i>	0.021* (2.17)	0.020* (1.94)	0.016 (0.81)	0.022 (1.15)	0.111*** (3.77)	0.114*** (3.73)	0.060 (0.96)	0.094 (1.41)
<i>Market-to-book</i>	-0.000 (-0.10)	-0.003 (-0.73)	-0.020** (-3.12)	-0.020** (-3.20)	-0.011 (-0.34)	-0.024 (-0.67)	-0.016 (-0.62)	-0.017 (-0.67)
<i>Return-on-assets</i>	-0.217* (-1.95)	-0.255* (-2.15)	-0.067 (-1.12)	-0.085 (-1.37)	-0.434** (-2.25)	-0.447** (-2.13)	-0.063 (-0.57)	-0.077 (-0.74)
<i>Leverage</i>	0.080** (2.42)	0.080** (2.35)	-0.070 (-1.04)	-0.063 (-0.92)	0.322* (1.77)	0.306 (1.53)	-0.021 (-0.08)	0.058 (0.20)
<i>Sales growth</i>	0.026 (0.74)	0.029 (0.90)	0.016 (1.28)	0.017 (1.40)	0.153 (1.57)	0.184* (1.67)	0.029 (0.50)	0.070 (1.12)
<i>Abnormal returns</i>	0.016 (0.68)	0.004 (0.17)	0.012 (0.91)	0.007 (0.48)	0.027 (0.34)	-0.008 (-0.10)	0.095** (2.38)	0.053 (1.29)
<i>Stock ret. volatility</i>	0.283 (1.82)	0.307* (1.93)	0.120 (0.16)	0.153 (0.20)	0.299 (0.40)	0.288 (0.39)	2.746 (0.94)	3.050 (1.03)
<i>Operating loss</i>	0.009 (0.51)	0.011 (0.61)	0.010 (0.52)	0.011 (0.56)	0.076 (1.47)	0.085* (1.74)	0.066 (0.66)	0.051 (0.52)
<i>CD&A exempt</i>	0.047 (0.39)	0.095 (0.79)	-0.046 (-0.45)	-0.019 (-0.19)	-0.030 (-0.14)	0.137 (0.58)	-0.456 (-1.46)	-0.361 (-1.21)
<i>ln(analysts)</i>	0.024 (1.48)	0.021 (1.22)	0.016 (1.36)	0.015 (1.27)	0.076 (1.63)	0.052 (1.08)	0.092** (2.17)	0.077* (1.76)
<i>Inst. ownership</i>	0.031 (0.57)	0.039 (0.71)	-0.007 (-0.25)	-0.007 (-0.23)	0.296** (2.21)	0.377*** (2.88)	-0.177 (-1.60)	-0.135 (-1.13)
<i>CEO turnover</i>	0.008 (0.34)	0.011 (0.50)	0.007 (0.92)	0.009 (1.11)	-0.010 (-0.19)	-0.016 (-0.29)	0.028 (0.94)	0.029 (0.93)
<i>ln(CEO equity pay)</i>	0.036* (2.30)	0.047*** (3.42)	0.000 (0.04)	0.007 (0.87)	0.141*** (3.66)	0.167*** (4.99)	0.001 (0.05)	0.032 (1.20)
<i>ln(CEO noneq. pay)</i>	0.039** (3.31)	0.041*** (3.49)	0.003 (0.32)	0.000 (0.04)	0.054* (1.83)	0.049* (1.78)	-0.021 (-0.53)	-0.043 (-1.01)
<i>Board independence</i>	0.067 (0.97)	0.066 (1.00)	-0.003 (-0.04)	-0.000 (-0.01)	0.620 (1.48)	0.547 (1.41)	0.025 (0.07)	-0.036 (-0.10)
<i>CC tenure</i>	-0.006*** (-4.74)	-0.006*** (-4.35)	0.003 (1.54)	0.003 (1.64)	-0.018*** (-2.74)	-0.015** (-2.01)	0.023*** (2.96)	0.025*** (3.27)
<i>Classified board</i>	-0.049** (-3.10)	-0.048** (-3.00)	-0.008 (-0.32)	-0.010 (-0.40)	-0.140** (-2.47)	-0.155*** (-2.74)	0.032 (0.29)	-0.003 (-0.03)
<i>Dual class shares</i>	-0.061 (-1.61)	-0.108** (-2.55)	0.013 (0.22)	0.018 (0.31)	-0.374*** (-3.47)	-0.584*** (-5.33)	0.097 (0.57)	0.139 (0.68)
Regression	OLS	OLS	OLS	OLS	Poisson	Poisson	Poisson	Poisson
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	No	No	Yes	Yes	No	No
Firm FE	No	No	Yes	Yes	No	No	Yes	Yes
Firm years	8,985	8,981	8,803	8,799	8,985	8,981	5,705	5,701
Adjusted (pseudo)-R ²	.160	.145	.668	.665	.183	.172	.420	.415

This table presents regression analyses of the determinants of shareholder engagement. We estimate regressions of *engagement indicator* (columns 1–4) and *engagement count* (columns 5–8) using Equation (1) for observations with nonmissing values for all determinants. All regressions include fiscal year fixed effects (FEs). Columns 1, 2, 5, and 6 use industry FEs based on one-digit Standard Industrial Classification codes and have standard errors double clustered by industry and year. Columns 3, 4, 7, and 8 use firm FEs with standard errors double clustered by firm and year. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

Across many specifications, *size*, *leverage*, and CEO pay tend to be positively related to engagement. Similarly, there is evidence that *analyst following*, *institutional ownership*, and *abnormal returns* are positively related to engagement at the 5% level in some regressions. Conversely, firms with *classified boards*, *dual class shares*, higher *market-to-book*, and higher profit (*return-on-assets*) tend to disclose less engagement. *Compensation committee tenure* is negatively related to engagement with industry FEs, but positively related to engagement count with firm FEs. Thus, in the cross section, higher compensation committee tenure appears to weaken monitoring. But, within a firm, once a compensation committee begins engaging, it likely continues to do so. *Board independence*, *stock return volatility*, *sales growth*, *operating losses*, *CD&A exemption*, and *CEO turnover* lack or have significance below the 5% level in all regressions. We exclude them as controls in our regressions of engagement in Section 5 to avoid overspecification issues (Whited et al. 2022). The results are not sensitive to this choice.

4. ISS and Shareholder Engagement

The results in Tables 2 and 3 suggest a strong relation between ISS recommendations, SOP voting, and shareholder engagement. In this section, we use an identification strategy to determine whether ISS's policies *cause* increases in engagement. We first explain and validate the identification strategy before using regression techniques to show the causal link between ISS and engagement.

4.1 Identification strategy

ISS considers firms with less than 70% SOP support as problematic and reviews their responsiveness to shareholder concerns. Firms with SOP voting support just below 70% receive ISS treatment, while those above do not. Thus, our identifying assumption is that whether a firm receives ISS treatment is locally random around this threshold.

We consider whether the 70% threshold is specific to ISS or is also a significant threshold for other key constituents. GL, the second-largest proxy advisory firm in the US, considers SOP results below 75% support from 2011 to 2017 and below 80% from 2018 onward as a trigger in its policies for encouraging shareholder engagement.¹⁶ Thus, GL's treatment encompasses our treated and control firms near ISS's 70% cutoff and should not drive our results.

Large institutional shareholders might also deem 70% to be an important threshold. To shed light on this question, we take two approaches. First, we examine survey results from ISS from 2011, when they established their

¹⁶ See Section 1.1 of the Internet Appendix for additional discussion and tests of Glass Lewis's policies.

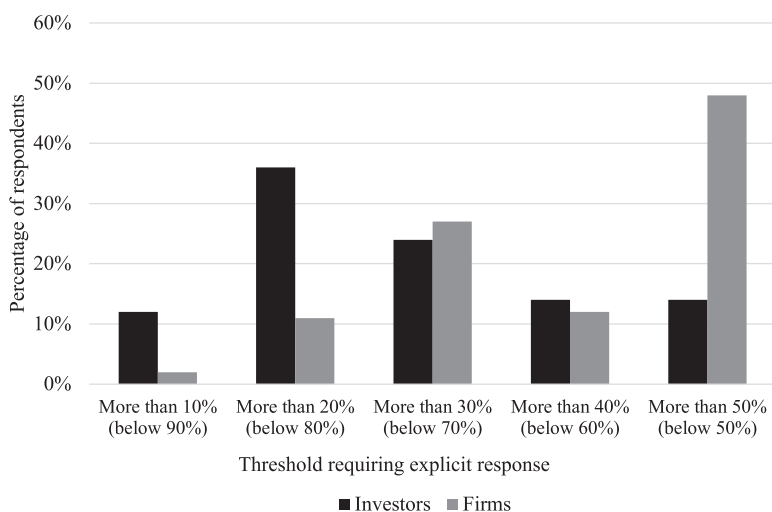


Figure 3
ISS survey on shareholder engagement

This figure charts investor and firm responses to a say-on-pay (SOP) survey conducted by Institutional Shareholder Services (ISS) in 2011. The survey asked investors and firms, “At what level of opposition on a say-on-pay proposal should there be an explicit response from the board regarding improvements on pay practices?” We tabulate responses across five thresholds on the *x*-axis and graph the percentage of investors and firms responding with support for each SOP voting threshold on the *y*-axis. See [ISS \(2011\)](#).

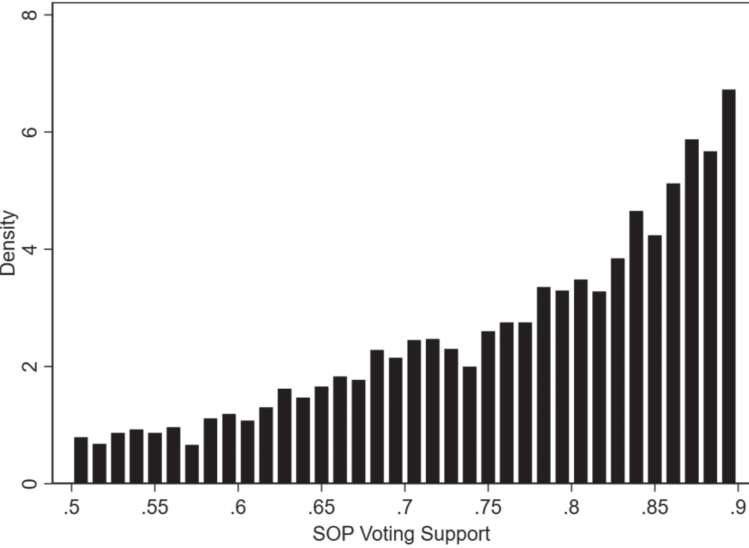
shareholder engagement policy. As shown in Figure 3, ISS asked both investors and firms at what SOP level should the board explicitly respond to issues related to pay practices. The possible choices were 50%, 60%, 70%, 80%, and 90%.

As the graph in Figure 3 depicts, the most common response from investors was 80%, while firms most frequently selected 50%. Thus, the figure depicts the divergence in investors’ and firms’ views and clarifies that the 70% threshold selected by ISS lacked unanimous support between market participants.

Second, previous research has shown that large investors (e.g., BlackRock) engage with portfolio firms on governance issues ([Pawliczek, Skinner, and Wellman 2021](#)). To help rule out that these investors’ engagement policies drive our results, we show in Table IA-6 of the Internet Appendix that our engagement results are not driven by ownership of the “Big 3” asset managers (i.e., BlackRock, State Street, and Vanguard). We also examine the stated policies of BlackRock, State Street, and Vanguard and find no language requiring portfolio firms to engage after a certain level of SOP dissent.

Next, we examine SOP voting outcomes for the full sample of 20,610 firm years to test whether receiving ISS treatment is random near the 70% threshold. Figure 4 plots the distribution of *SOP voting support* for firms with support between 50% and 90% (panel A) and 67.5% to 72.5% (panel B). The distribution shows no visual clustering around the 70% threshold in either panel, providing initial evidence that treatment is locally random.

A SOP voting support between 50% and 90% for annual meetings over 2011 to 2019



B SOP voting support between 67.5% and 72.5% for annual meetings over 2011 to 2019

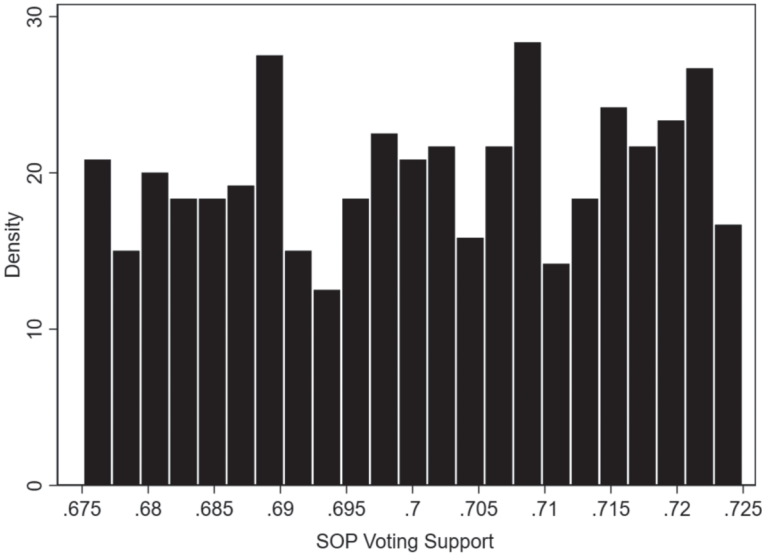


Figure 4
Distribution of say-on-pay voting support

This figure presents a histogram of the percentage of shareholders approving say-on-pay (SOP) using the ISS definition of *SOP voting support*. Panel A presents the density of firms with voting support between 50% and 90%, while panel B shows the density of firms with voting support between 67.5% and 72.5%.

We then conduct a formal test to check for manipulation of the assignment variable around the 70% cutoff (McCrary 2008). Specifically, we use the local polynomial density estimation of Cattaneo, Jansson, and Ma (2020) (hereinafter “CJM”), who develop a robust nonparametric density estimator to test for manipulation in the assignment variable in an RD setting. These tests are based on the key underpinnings of a valid RD design in that treated firms should not be able to precisely control the assignment variable around the deterministic cutoff. Thus, the results should show that the marginal density in *SOP voting support* is continuous for our sample around the 70% cutoff.

In panel A of Table 4, we test for discontinuities in *SOP voting support* around the 70% threshold for different bandwidths ($\pm 2.5\%$, $\pm 5.0\%$, $\pm 10\%$) over the full sample period. We also examine the data-driven bandwidths developed by CJM for the full sample period and for the first year of SOP voting in 2011. We report the robust bias-corrected manipulation test statistic, *p*-value, and the number of treated and control firms in each range. A statistically significant manipulation test statistic would suggest that managers can successfully influence the assignment variable (*SOP voting support*) around the 70% threshold, while a nonsignificant statistic would indicate no detected manipulation. These results are used to determine whether the assignment of ISS treatment is random.

Using local linear regressions, we find no evidence of a discontinuity in *SOP voting support* around the 70% threshold. The robust bias-corrected manipulation test statistic is not statistically different from zero in any of the bandwidths, thereby supporting the visual inferences in Figure 4. Therefore, we find no evidence that firms cluster just above the 70% threshold to avoid ISS scrutiny.¹⁷

Our third validation exercise tests for differences in many preassignment characteristics between ISS-treated and control firms before the SOP vote. Table 4 shows the average values for these characteristics for firms within a range of 2.5% above or below the 70% SOP voting support threshold. Columns 1 and 2 show the mean values for ISS-treated and control firms, respectively. Column 3 reports the difference between the means, and column 4 shows the *p*-value for the difference being statistically different from zero. Column 5 presents the number of treated and control firm observations for each test.

The variables in this validation exercise can be grouped into 7 categories: SOP voting support, shareholder engagement, firm characteristics, CEO pay, director voting, governance, and board characteristics. Appendix C describes all the variables. We compare certain director characteristics, such as the

¹⁷ Using a sample period that ends in 2016, Bach and Metzger (2019) find evidence of voting manipulation around the 50% threshold. Importantly, they test ISS's 70% threshold for SOP voting support and also find no evidence of manipulation. Specifically, they state that SOP “vote manipulation only takes place at the official threshold of 50%” (Bach and Metzger 2019). In Table IA-7 of the Internet Appendix, we replicate their findings after extending the sample through 2019. We confirm that evidence of manipulation exists around the 50% SOP voting support threshold, but not the 70% threshold.

Table 4
Validity of research design

A. No manipulation of forcing variable

	(1)	(2)	(3)	(4)	(5)
Bandwidth	±2.5%	±5.0%	±10.0%	CJM	CJM
Manipulation test statistic	0.537 (0.592)	0.634 (0.526)	-0.514 (0.608)	0.765 (0.444)	-0.262 (0.794)
Sample period	2011–2019	2011–2019	2011–2019	2011–2019	2011
Treated firms (below)	262	479	804	777	48
Control firms (above)	291	563	1,273	1,096	26

B. Similarity of treatment and control sample

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Treated	Control	Diff.	p-value	RD Treated	RD p-value	N [T:C]
Say-on-pay voting							
SOP voting support _{t-1}	0.837	0.844	-0.007	.702	0.057	.301	155:162
SOP voting support _t	0.688	0.713	-0.025***	.000	na	na	209:217
Shareholder engagement_t							
Engagement indicator	0.273	0.263	0.010	.815	-0.097	.520	209:217
Engagement count	0.986	1.097	-0.111	.630	-0.119	.638	209:217
Engagement ref. SOP	0.234	0.249	-0.014	.729	-0.119	.403	209:217
Engagement table	0.100	0.074	0.027	.328	-0.060	.533	209:217
Shareholders contacted	0.034	0.027	0.007	.607	-0.060	.304	159:169
Shareholders spoken with	0.051	0.053	-0.002	.892	-0.034	.575	169:181
Shareholder engagement_{t-1}							
Engagement indicator	0.146	0.162	-0.016	.662	-0.142	.178	205:210
Engagement count	0.429	0.529	-0.099	.448	-0.189	.184	205:210
Engagement ref. SOP	0.127	0.133	-0.007	.844	-0.052	.661	205:210
Engagement table	0.049	0.062	-0.013	.560	-0.063	.362	205:210
Shareholders contacted	0.037	0.016	0.021	.116	-0.062	.163	186:182
Shareholders spoken with	0.025	0.028	-0.003	.801	-0.019	.614	186:188
Firm characteristics_t							
ISS "against" SOP	0.689	0.650	0.039	.391	-0.066	.676	209:217
Size	7.303	7.220	0.083	.676	0.063	.918	209:217
Market-to-book	1.478	1.519	-0.040	.839	0.254	.663	209:217
Return-on-assets	0.005	0.000	0.005	.830	0.038	.401	209:217
Leverage	0.221	0.224	-0.003	.884	-0.079	.295	209:217
Sales growth	0.067	0.079	-0.012	.842	-0.185	.219	209:217
Abnormal returns	-0.068	-0.072	0.004	.911	0.152	.123	209:217
Stock return volatility	0.124	0.135	-0.011	.157	-0.039	.060*	209:217
Operating loss	0.641	0.677	-0.036	.431	-0.144	.336	209:217
CD&A exempt	0.072	0.083	-0.011	.667	-0.124	.178	209:217
ln(analysts)	1.619	1.572	0.047	.658	0.006	.985	209:217
Inst. ownership	0.615	0.617	-0.002	.947	-0.124	.250	209:217
CEO turnover	0.134	0.161	-0.027	.428	0.072	.392	209:217
Adjusted stock returns	-0.119	-0.126	0.007	.826	0.098	.297	209:217
Adjusted ROA	-0.012	0.004	-0.017	.441	0.033	.614	209:217
Net income (\$B)	0.285	0.246	0.039	.613	0.117	.740	209:217
CEO pay_t							
ln(CEO equity pay)	12.887	13.331	-0.444	.303	1.764	.219	209:217
ln(CEO noneq. pay)	13.948	14.209	-0.262	.172	0.174	.603	209:217
Expected CEO pay	14.887	15.036	-0.149	.198	0.396	.213	209:217
Unexpected CEO pay	-0.058	0.058	-0.116	.406	-0.021	.948	209:217
Director voting							
CC member support _t	0.892	0.895	-0.003	.778	0.010	.761	183:197
CC member support _{t-1}	0.909	0.909	0.000	.980	-0.017	.576	176:181
CC chair support _t	0.892	0.905	-0.014	.244	0.008	.856	142:143
CC chair support _{t-1}	0.916	0.901	0.015	.268	-0.018	.607	126:136
Director support _t	0.909	0.911	-0.002	.821	-0.001	.964	209:217
Director support _{t-1}	0.921	0.921	0.000	.959	-0.030	.316	204:214

Table 4
(Continued)

B. Continued

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Treated	Control	Diff.	<i>p</i> -value	RD Treated	RD <i>p</i> -value	N [T:C]
Governance_{<i>t</i>}							
<i>CC tenure</i>	8.527	8.896	−0.369	.404	−2.400	.080*	209:217
<i>Classified board</i>	0.421	0.433	−0.012	.801	0.036	.809	209:217
<i>Board independence</i>	0.793	0.783	0.010	.364	0.039	.239	209:217
<i>Dual class shares</i>	0.043	0.046	−0.003	.880	−0.004	.963	209:217
Board characteristics_{<i>t+1</i>}							
<i>Classified board</i>	0.402	0.419	−0.017	.715	0.039	.793	209:217
<i>Board size</i>	8.440	8.585	−0.145	.546	−0.647	.324	209:217
<i>Directors voted on</i>	6.230	6.346	−0.116	.737	−0.540	.582	209:217
<i>CC members voted on</i>	2.474	2.548	−0.075	.604	−0.371	.413	209:217
<i>CC chair voted on</i>	0.675	0.654	0.020	.659	0.070	.581	209:217

Panel A reports tests of whether firms manipulate the forcing variable, *SOP voting support*, around the 70% threshold to avoid ISS treatment. We report the *manipulation test statistic* and corresponding *p*-values using the robust bias-corrected density estimator of Cattaneo, Jansson, and Ma (2020) (“CJM”) to test for discontinuities for various bandwidths and sample periods around the 70% cutoff. Panel B compares differences in say-on-pay (SOP) voting, shareholder engagement, firm characteristics, CEO pay, director voting, governance, and board characteristics during the preassignment period (i.e., prior to receiving 67.5% to 72.5% SOP voting support in year *t*) or the treatment period (i.e., year *t* + 1). *Treated* includes firms with 67.50% to 69.99% SOP voting support. *Control* includes firms with 70.00% to 72.50% SOP voting support. In column 5, we present the robust bias-corrected coefficient for *Treated* using a local linear regression discontinuity estimation from Cattaneo, Jansson, and Ma (2018). The *p*-value from the RD estimation is reported in column 6. The number of treat (T) and control (C) observations are in column 7. We define variables in Appendix C. **p* < .1; ***p* < .05; ****p* < .01 (using two-tailed *t*-tests and the reported *p*-value in column 4).

presence of a classified board and how many directors are voted on for the year *t* + 1, as they are known in advance and could potentially influence engagement. We compare most other variables in year *t* just before the low SOP vote. For many variables, we also examine the year *t* − 1 values (e.g., SOP voting support, engagement, and director voting) to ensure there are no trends that span for more than 1 year.

While most of the preassignment variables are commonly used in governance literature, we also consider whether unexpectedly high CEO pay might affect engagement. *Expected CEO pay* is the estimate of the CEO’s total compensation using the regression methodology in Rouen (2020). *Unexpected CEO pay* is calculated as actual CEO total compensation less *expected CEO pay*. The results in panel B of Table 4 indicate that there are no significant univariate differences between firms just below or above the 70% threshold, except for the assignment variable (*SOP voting support*).

We also use local linear RD estimates to test for differences in preassignment variables, following the robust bias-corrected method proposed by Calonico et al. (2017), whose work we will describe further in the next subsection. The RD estimates and *p*-value, shown in columns 5 and 6, indicate that there are practically no significant differences between firms just below and just above the 70% threshold for most dimensions. The RD estimate for *stock return volatility* is slightly negative and marginally significant at the 10% level, suggesting that ISS-treated firms have lower return volatility. But earlier

analysis in Table 3 shows that return volatility is not related to engagement in the overall universe of firms in the ISS database, so this difference is not expected to bias the results. We find that differences in preassignment *CC tenure* are marginally significant at the 10% level in the RD test. However, further examination reported in Figure IA-1 of the Internet Appendix shows that this result is driven by outliers on each side of the 70% threshold.

Taken together, the results in Table 4 and Figure 4 support the notion that the distribution of firms around the 70% threshold for ISS treatment is largely random. We find no evidence that firms manipulate their SOP voting support to avoid treatment and the attributes of firms around the threshold are similar, which allows us to link changes in shareholder engagement to the effect of ISS's policy.

4.2 Research design

To test the effect of ISS treatment on engagement, we consider the SOP vote by shareholders of firm i at time t , the annual meeting date. The percentage of votes approving SOP is the vote share v_{it} . If v_{it} is below the 70% vote threshold, v^* , then it receives ISS treatment, and we code the indicator for treatment as $ISS\ treatment_{it} = 1$. When $v_{it} \geq v^*$, $ISS\ treatment_{it}$ takes the value of 0.

We estimate the effect of ISS treatment on the outcome variables using both an average treatment effect (ATE) estimation and an RD approach. The ATE estimation tests use this equation:

$$y_{it+1} = \kappa + \theta\ ISS\ treatment_{it} + X_{it} + \varphi_j + \mu_t + \varepsilon_{it}, \quad (2)$$

where the coefficient of interest θ is the effect of ISS on the outcome variable y_{it+1} (e.g., *engagement indicator*) after a low SOP vote. We include industry FEs (φ_j) using one-digit SIC codes, which controls for time-invariant unobserved heterogeneity across industries, and fiscal year FEs (μ_t) to control for time trends that affect all firms. X_{it} is a vector of firm-level control variables identified as determinants of engagement in Table 3, including *size*, *market-to-book*, *return-on-assets*, *leverage*, *abnormal returns*, *ISS "against" SOP*, *ln(analysts)*, *institutional ownership*, *classified boards*, *dual class shares*, *compensation committee tenure*, and *director voting support*.

These tests are based on two key assumptions. The first is that, without the influence of ISS, the engagement of treated and control firms would have followed similar patterns (i.e., the parallel trends assumption). We examine this assumption in Section 5.5.

The second assumption is the stable unit treatment value assumption (SUTVA). SUTVA holds that for an estimation to provide unbiased estimates of the average treatment effect, the impact on treated firms should not influence the outcomes of control firms (e.g., Berg, Reisinger, and Streitz 2021). In our setting, control firms might learn from the response of ISS-treated firms, potentially reducing the cost of initiating their own engagement activities. These firms might also share similar compensation consultants or board

members, a situation that could lead to knowledge spillovers on how to respond to a low SOP vote. Both situations, however, would lead us to *underestimate* the ATE of ISS's policy on engagement by increasing the level of engagement of the control firms.

We also use an RD analysis to study the effect of ISS on the primary engagement measures: *engagement indicator* and *engagement count*. We use a Sharp RD estimate that does not control for covariates or FEs because Table 4 shows almost no significant differences in just under 50 preassignment covariates in treatment and control firms, and Figure 4 and Table 4 show no discontinuities in the assignment variable around the 70% threshold. We employ the robust RD method of Calonico et al. (2017) that adjusts the confidence intervals and variance estimator to improve the accuracy of the coefficient estimates and statistical significance. For all tests, we report robust bias-corrected RD estimates.

To ensure the robustness of our RD specification, we test ISS treatment using different bandwidths around the 70% threshold (2.5%, 5.0%, and 10.0%) and the data-driven bandwidth selection method of Calonico, Cattaneo, and Farrell (2020) ("CCF"). We primarily use a local linear RD estimate, as higher order polynomials can introduce bias (Gelman and Imbens 2019), but also report a local quadratic result for robustness. We use a triangular kernel to construct all local-polynomial estimators.

4.3 Engagement in the year after ISS treatment

This subsection examines how ISS affects shareholder engagement in the year following treatment. We first provide summary statistics and then conduct the regression analysis to test H1 using a subset of firms whose SOP votes fall within $\pm 2.5\%$ of 70% support. We also use varying bandwidths around the 70% threshold for the RD analysis.

Panel A of Table 5 shows the average, median, and standard deviation of engagement measures for ISS treated and control firms that had SOP votes within $\pm 2.5\%$ of the 70% support threshold. In the year after receiving a low SOP vote, 58.5% of the sample firms report shareholder engagement, on average using 2.1 engagement words. ISS treated firms are 14.9-percentage-points more likely to report shareholder engagement and use 0.74 more engagement words than control firms, both of which are significant at the 1% level or better using two-tailed *t*-tests. These differences are economically large and reflect a 25% and 36% increase, respectively, versus the mean values for treated and control firms.

Approximately 52% of the sample firms report engaging with investors in response to the previous year's low SOP vote. About 14% provide a table detailing shareholders' concerns and the board's response. On average, sample firms contact shareholders owning 17% of shares and report speaking with owners of 12% of outstanding shares. When only considering firms that disclose engagement, they contact shareholders holding an average of 58% of

Table 5

ISS treatment and shareholder engagement

A. Summary statistics and univariate tests

Shareholder engagement _{t+1}	(1)	Treated		(3)	(4)	Control		(6)	(7)	(8)	(9)	(10)
	Mean	Median	SD	Mean	Mean	Median	SD	Mean	p-value	N [T:C]	Mean	
Engagement indicator	0.660	1	0.475	0.512	1	1	0.501	0.149***	.002	209:217	0.585	
Engagement count	2.464	1	3.200	1.724	1	1	2.624	0.741***	.009	209:217	2.087	
Engagement references SOP	0.598	1	0.491	0.442	0	0	0.498	0.156***	.001	209:217	0.519	
Engagement table	0.191	0	0.394	0.092	0	0	0.290	0.099***	.003	209:217	0.141	
Shareholders contacted	0.237	0	0.316	0.103	0	0	0.213	0.134***	.000	115:132	0.166	
Shareholders contacted(nonzero)	0.619	0.645	0.149	0.505	0.500	0.500	0.137	0.115***	.002	44:27	0.576	
Shareholders spoken with	0.172	0	0.235	0.078	0	0	0.172	0.094***	.000	118:130	0.123	
Shareholders spoken with(nonzero)	0.431	0.454	0.159	0.405	0.430	0.430	0.148	0.026	.492	47:25	0.422	

B. ISS treatment and shareholder engagement_{t+1}

Dependent variable	(1)	(2)	(3)	(4)	(5)	(6)
	Engagement indicator	Engagement count	Engagement references SOP	Engagement table	Shareholders contacted	Shareholders spoken with
ISS treatment	0.160*** (3.78)	0.407*** (3.57)	0.165*** (3.88)	0.100*** (3.09)	0.126*** (4.40)	0.081*** (3.59)
Regression	OLS	Poisson	OLS	OLS	OLS	OLS
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year, industry FE	426	426	426	426	247	248
Firms						
Adjusted (pseudo)-R ²	.251	.226	.267	.120	.368	.337

This table reports tests of the relation between ISS treatment and subsequent shareholder engagement in year $t+1$. ISS treatment equals one if the firm has say-on-pay (SOP) voting support in between 67.50% and 69.99%, and zero if SOP voting support is between 70.00% and 72.50%. Panel A reports the mean, median, and standard deviation of engagement values for treated firms in columns 1–3 and control firms in columns 4–6. We tabulate the mean difference in each variable and the corresponding p -value from a two-tailed t -test in columns 7 and 8. Column 9 shows the count of treated (T) and control (C) firms. As a point of reference, column 10 displays the combined averages for treated and control firms (T+C). Panel B presents regressions of the average treatment effect of ISS on shareholder engagement using estimates of Equation (2). Columns 5 and 6 exclude firms that report engagement but do not provide the percentage of shareholders contacted or spoken with. The number of observations differs as some firms only report one of these two metrics. All regressions include fiscal year and industry fixed effects. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

shares and spoke with holders of 42% of shares. ISS treated firms engage with shareholders holding an average of 13.4% more shares (23.7% vs. 10.3%) and spoke with investors holding 9.4% more shares (17.2% vs. 7.8%) compared to control firms. Both differences are significant at the 1% level and reflect an 81% and 76% increase versus the sample mean, respectively.

In panel B, we study the relation between ISS treatment and engagement. We estimate Equation (2) using OLS regressions for most of the dependent variables and a Poisson regression for the *engagement count*. The results show that *ISS treatment* is positively related to the presence and intensity of shareholder engagement in the following year. In all six regressions, the coefficient for *ISS treatment* is statistically significant at the 1% level and the size of the engagement differences is substantial. For example, treated firms are 16.0-percentage-points more likely to report engagement (column 1), which is a 27% increase from the sample mean of 58.5%. The coefficient for *ISS treatment* in column 2 is 0.407, indicating that the *engagement count* for treated firms is 50% higher.¹⁸ In columns 3 to 6, we find similar results showing a positive and significant relation between ISS treatment and various measures of the depth of shareholder engagement.

Table 6 reports the RD analysis of engagement. It includes the RD estimate, *p*-values, number of firms in the ISS treated and control sample for each bandwidth, and polynomial type. Panel A presents the results for the *engagement indicator* as the dependent variable, showing that ISS treatment leads to a significantly higher probability of engagement in all five RD specifications.

Our RD estimates show that treatment significantly increases the probability of engagement by 19.7 to 33.4 percentage points. These increases are larger than the ATE estimations using OLS in Table 5, possibly because our RD approach uses a triangular kernel, which gives more weight to observations near the 70% threshold (Cattaneo and Titiunik 2022). For example, the narrow bandwidth of 2.5% in column 1 estimates a 33.4-percentage-point increase in engagement for ISS-treated firms, while the broader bandwidth of 10% in column 3 results in a 19.7-percentage-point increase for treated firms. The data-driven bandwidth in columns 4 and 5 estimates a 16.5- to 21.7-percentage-point increase in engagement for ISS-treated firms. All these results are statistically significant and directionally consistent with the ATE analysis in Table 5.

Table 6, panel B, shows the results of using the same RD specifications with *engagement count* as the dependent variable. To account for the skewed distribution due to zero engagement, we use a log-transformed value of one plus the *engagement count* as the dependent variable. The five specifications all suggest that ISS treatment increases the intensive margins of engagement.

Figure 5 shows the discontinuity in engagement with ISS treatment through RD plots of the *engagement indicator* (panel A) and *engagement count* (panel B). Each circle in the plot represents an evenly spaced bin of

¹⁸ This difference is calculated by exponentiating the *ISS treatment* coefficient: $e^{(0.407)} - 1 = 0.502$.

Table 6
Regression discontinuity tests of ISS treatment and shareholder engagement

A. Regression discontinuity tests of shareholder engagement indicator_{t+1}

	(1)	(2)	(3)	(4)	(5)
Bandwidth:	±2.5%	±5.0%	±10.0%	CCF	CCF
<i>ISS treatment</i>	0.334*** (.010)	0.290*** (.002)	0.197*** (.003)	0.165*** (.007)	0.217*** (.002)
Polynomial Kernel	Linear Triangular	Linear Triangular	Linear Triangular	Linear Triangular	Quadratic Triangular
Treated firms (below)	262	479	804	639	814
Control firms (above)	291	563	1,273	849	1,299

B. Regression discontinuity tests of shareholder engagement count_{t+1}

	(1)	(2)	(3)	(4)	(5)
Bandwidth:	±2.5%	±5.0%	±10.0%	CCF	CCF
<i>ISS treatment</i>	0.446** (.029)	0.345** (.019)	0.298*** (.005)	0.269*** (.006)	0.323*** (.004)
Polynomial Kernel	Linear Triangular	Linear Triangular	Linear Triangular	Linear Triangular	Quadratic Triangular
Treated firms (below)	262	479	804	628	849
Control firms (above)	291	563	1,273	833	1,378

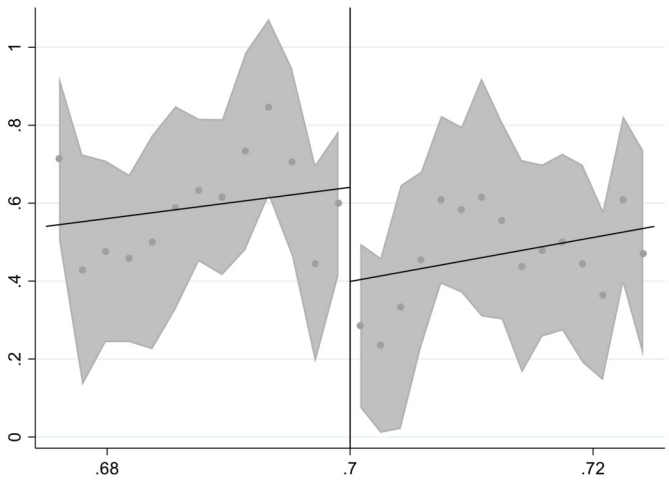
This table presents local polynomial regression discontinuity (RD) estimates of shareholder engagement using the “rdrobust” package in Stata from [Cattaneo, Jansson, and Ma \(2018\)](#). We use a triangular kernel to construct all local-polynomial estimators. Panel A tests the *engagement indicator*. Panel B tests the natural logarithm of one plus *engagement count*. We present local linear robust bias-corrected RD estimates for the 2.5%, 5.0%, and 10.0% bandwidth around the *ISS treatment* threshold of 70% say-on-pay (SOP) voting support in columns 1, 2, and 3, respectively. Columns 4 and 5 utilize the bandwidth selection procedure in [Calónico, Cattaneo, and Farrell \(2020\)](#) (“CCF”). We present these results for both a local linear (column 4) and local quadratic (column 5) polynomial. We report the number of treated and control firms for each estimation, which are those just below and above the 70% SOP voting support threshold, respectively. * $p < .1$; ** $p < .05$; *** $p < .01$ (based on the corresponding z-statistic and reported p -value in parentheses).

observations with a bandwidth of $\pm 2.5\%$ around the 70% threshold. The gray-shaded region shows the 95% confidence interval, and the solid line fits a first-order polynomial to the relation between SOP voting support and shareholder engagement.

The graphs in Figure 5 illustrate that firms receiving ISS treatment just below the 70% threshold have a noticeable increase in engagement compared to control firms just above it. The control firms, on the other hand, have a noticeable decline in engagement when they just miss treatment and are in the bins closest to 70% SOP voting support. There is slightly more variation in engagement among the control firm bins, with a greater number of bins plotted closer to zero engagement.

In Table IA-8 of the Internet Appendix, we also examine whether ISS-treated firms are more inclined to reference ISS recommendations or their SOP policies in firms’ SEC filings in the year after falling below the 70% threshold. Using the SEC’s EDGAR Full Text search tool, we find that ISS-treated firms are 54% more likely to reference ISS in its filings than control firms. This evidence bolsters the conclusion that ISS plays an important role in firms’ engagement decisions.

A Shareholder engagement indicator



B Shareholder engagement count

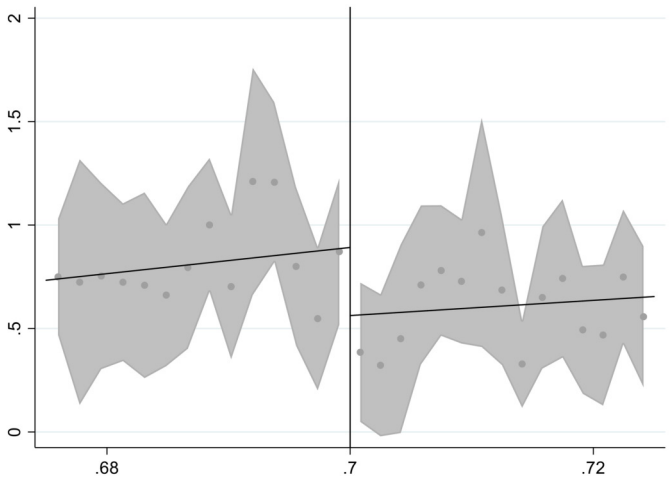


Figure 5
Regression discontinuity plots of shareholder engagement

This figure presents the regression discontinuity plots of the *Engagement indicator* (panel A) and *Engagement count* (panel B) in year $t+1$. Each circle depicts an evenly spaced bin with a bandwidth of $\pm 2.5\%$ around the 70% say-on-pay (SOP) voting support threshold. ISS-treated firms have SOP voting support between 67.50% and 69.99%, while control firms have SOP voting support between 70.00% and 72.50%. The gray-shaded region depicts the 95% confidence interval. The solid line fits a first-order polynomial to estimate the relation between SOP voting support and shareholder engagement. We define variables in Appendix C.

Overall, the results in Tables 5 and 6, and Figure 5 strongly support the idea that ISS treatment prompts firms to engage more with their shareholders when SOP voting support falls just below the 70% threshold. We interpret this collective evidence as providing strong causal support for H1.

4.4 Shareholder engagement and ISS sanctions

4.4.1 Ex ante director voting support In this subsection, we test whether firms whose board members have greater incentives to avoid a negative recommendation from ISS are more responsive to ISS treatment. H2a posits that boards with a greater threat of ISS sanctions, such as those with directors who had previously received lower shareholder voting support, will demonstrate a stronger engagement response to avoid ISS opposition to their reelection.

To test this, we modify Equation (2) to include controls for *ex ante compensation committee (CC) member voting support* and *director voting support*, which reflect the previous year's voting support levels. We also include the interactions between these two voting support metrics and *ISS treatment*. The dependent variables are *engagement indicator* and *engagement count*. The variable of interest in each regression is *ISS treatment* $\times$ *voting support*. If firms facing a greater risk of sanctions under ISS treatment intensify engagement efforts more, we expect to see a negative coefficient for this interaction term.

In Table 7, the coefficient for *voting support* is positive and significant at either the 5% or 10% level in all regressions, indicating a relation between ex ante voting support and engagement. For tests of the interaction term, *ISS treatment* $\times$ *voting support*, we find no evidence that ISS treatment influences the extensive margin of engagement (columns 1 and 2). However, in testing for the intensive margin of engagement in columns 3 and 4, we find significant negative coefficients for the interaction terms for both director-voting based measures. This evidence suggests that while directors who are more susceptible to ISS sanctions (i.e., lower prior voting support) are not significantly more likely to engage, they do so more intensively, likely to demonstrate a sufficiently robust response in accordance with ISS's policies. Thus, our findings offer support for H2a, indicating that the threat of ISS opposing their reelection prompts directors to adhere to ISS's engagement policies in response to a low SOP vote.

4.4.2 Ex post ISS recommendations Our hypotheses and inferences above rely on (1) the credibility of ISS's threat of recommending against the directors of nonresponsive treated firms, and (2) directors being responsive to avoid the negative recommendations. In this subsection, we formally test H2b, namely, that ISS will follow through with the threat to recommend voting against the directors of less responsive firms.

Table 7
Shareholder engagement and prior voting support

	(1)	(2)	(3)	(4)
	<i>Engagement indicator</i>		<i>Engagement count</i>	
	CC members	All directors	CC members	All directors
Voting support				
<i>ISS treatment</i>	0.480 (1.13)	0.254 (0.58)	3.188** (2.50)	3.593** (2.29)
<i>Voting support</i>	0.611* (1.90)	0.735** (2.22)	2.763** (2.57)	3.007** (2.46)
<i>ISS treatment × voting support</i>	−0.365 (−0.77)	−0.103 (−0.21)	−3.060** (−2.23)	−3.415** (−2.05)
Regression	OLS	OLS	Poisson	Poisson
Controls	Yes	Yes	Yes	Yes
Year, industry FE	Yes	Yes	Yes	Yes
Observations	380	426	380	426
Adjusted (pseudo)- R^2	.237	.249	.235	.231

This table examines whether directors facing a higher risk of ISS voting opposition engage more actively following ISS treatment. We estimate regressions of *engagement indicator* (columns 1 and 2) and *engagement count* (columns 3 and 4) after modifying Equation (2) to include proxies for ISS sanction risk. For these, we use two ex ante voting support variables: *compensation committee (CC) member voting support*, and *all directors voting support*. In this table, *ISS treatment* equals one if the firm has say-on-pay (SOP) voting support in between 67.50% and 69.99%, and zero if SOP voting support is between 70.00% and 72.50%. The variable of interest in each test is the ISS treatment interacted with the two voting support variables (*ISS treatment × voting support*), which indicates whether the engagement response to ISS treatment differs for firms with higher baseline values of each measure. All regressions include year and industry fixed effects. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

Because we do not have access to ISS’s proprietary evaluation of firm responsiveness, we classify firms as more or less responsive based on whether they disclose engagement in their subsequent proxy statements. Our dependent variables in these analyses are whether ISS recommends voting against compensation committee members and separately members of the full board after a low SOP vote. We again compare ISS-treated and control firms, categorized by whether they engage with shareholders. If the ISS threat is credible, we would expect less responsive treated firms to be more likely to receive “against” recommendations for their directors than more responsive treated firms. We also posit that ISS’s increased scrutiny will result in more negative voting recommendations for treated than control firms within the same category of responsiveness.

In our test, we designate firms as *more responsive* if they disclose engagement in year $t+1$ and *less responsive* if not. We then evaluate the likelihood that ISS recommends voting against board members in year $t+1$. If ISS’s threat is credible, we anticipate ISS would be more likely to recommend voting “against” directors of less responsive firms, especially those on the compensation committee. Additionally, because of increased scrutiny on treated firms, we expect that ISS would be more likely to recommend voting “against” directors of treated firms than directors of control firms with the same responsiveness.

Table 8
Credibility of the ISS threat

A. Likelihood of ISS recommending “against” directors in year after low vote

	(1)	(2)	(3)	(4)	(5)	(6)
	All	Less responsive	More responsive	Diff.	p-value	N [LR:MR]
CC member						
Treated	0.239	0.429	0.133	0.295***	.000	75:42
Control	0.081	0.115	0.052	0.063	.174	67:58
Difference	0.158***	0.314***	0.081*			
p-value	.001	.000	.091			
Director						
Treated	0.162	0.269	0.100	0.169***	.002	80:46
Control	0.077	0.099	0.057	0.042	.264	71:63
Difference	0.085***	0.170***	0.043			
p-value	.009	.003	.250			

B. Regressions of the likelihood of ISS recommending “against” directors in year after low vote

	(1)	(2)	(3)	(4)
	CC members		All directors	
ISS treatment	0.169*** (3.69)	0.063 (1.08)	0.086*** (2.66)	0.024 (0.57)
Less responsive	0.101* (1.90)	−0.028 (−0.40)	0.053 (1.43)	−0.021 (−0.43)
ISS treatment × less responsive		0.264*** (2.90)		0.153** (2.39)
Regression	OLS	OLS	OLS	OLS
Controls	Yes	Yes	Yes	Yes
Firm, industry FE	Yes	Yes	Yes	Yes
Observations	240	240	258	258
Adjusted R ²	.131	.161	.159	.176

This table presents results of tests to determine whether directors with lower engagement levels after ISS treatment are more likely to receive an “against” recommendation from ISS for their reelection in the following year. In this table, *ISS treatment* equals one if the firm has say-on-pay (SOP) voting support in between 67.50% and 69.99%, and 0 if SOP voting support is between 70.00% and 72.50%. We designate firms as *more responsive* if they disclose engagement in year $t+1$ and *less responsive* if they do not. Panel A presents the univariate likelihood of ISS recommending against compensation committee (CC) members and all directors in year $t+1$ based on the responsiveness of treated and control firms. Panel B presents regression results in tests that modify Equation (2) to include controls for the less responsive indicator and the interaction of treatment with responsiveness (*ISS treatment* × *less responsive*), which indicates whether the ISS’s voting recommendations differ for firms based on their engagement response to a low SOP vote. All regressions include year and industry fixed effects. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

Panel A of Table 8 reveals that while ISS recommends voting against 23.9% of treated directors on the compensation committee, this increases to 42.9% for less responsive firms, compared to 13.3% of more responsive firms. This 29.5-percentage-point difference (significant at the 1% level) is consistent with the threat of a negative ISS voting recommendation being credible for firms whose votes fall below the 70% threshold. In contrast, compensation committee members for less responsive control firms see an insignificant change in ISS recommendations. Further, column 2 shows that ISS more heavily penalizes treated firms that are less responsive, with a 31.4-percentage-point greater chance of an “against” recommendation for their compensation committee members compared to control firms.

The results are similar for the full board, though with smaller magnitudes. Directors of less responsive treated firms have a 16.9-percentage-point greater likelihood of receiving an “against” recommendation from ISS. This is consistent with ISS’s policy of opposing reelection of compensation committee members after an insufficient response to a low SOP vote before opposing the entire board.

In panel B of Table 8, we conduct OLS regression where the outcome variable equals one for an “against” director vote recommendation by ISS, and zero otherwise. We separately evaluate the compensation committee members and the full board. The independent variables include the *ISS treatment* and *less responsive* measures. The variable of interest is their interaction, *ISS treatment* $\times$ *less responsive*. A positive interaction coefficient would be consistent with ISS placing greater scrutiny on the responsiveness of treated versus control firms. We hold constant the same factors as in Equation (2) and include the same FEs.

The results in columns 1 and 3 show that ISS treatment increases the chance of an “against” vote recommendation for the members of the compensation committee and the entire board, highlighting ISS’s elevated scrutiny of treated firms. Less responsive firms are similarly more likely to receive “against” recommendations from ISS. Importantly, columns 2 and 4 show that firms that are both ISS-treated and less responsive have a 26.4- and 15.3-percentage-point increase in the likelihood of an “against” recommendation for the compensation committee members and all directors, respectively. The coefficients for both interaction terms are significant at the 5% level or better.

We interpret the above as compelling evidence that ISS’s potential sanctions are a credible threat for directors that do not strengthen shareholder outreach efforts following a low vote. Collectively, the evidence in support of H2a and H2b is consistent with our assumption regarding ISS’s credibility as well as strengthens the causal attribution of firms’ engagement responses to ISS in tests of H1.

4.5 Engagement over time after ISS treatment

Next, we examine the impact of ISS treatment on engagement over time. These tests have two goals. First, we want to assess the parallel trends assumption. Second, we wish to determine whether the initial increase in engagement after ISS treatment persists (H3). To do this, we measure engagement over a 9-year period centered on the low SOP vote and use indicator variables in an event-time d-i-d design to estimate dynamic treatment effects.

To formally test the impact of ISS on shareholder engagement over time, we first use a panel regression to estimate this equation:

$$\text{Engagement}_{it+1} = \kappa + \theta \text{ISS treatment}_{it} + \mathbf{X}_{it} + \omega_i + \mu_t + \varepsilon_{it}, \quad (3)$$

where the coefficient of interest θ is the effect of ISS on shareholder engagement. We test the presence (*engagement indicator*) and intensity

(*engagement count*) of engagement separately using a FE OLS and Poisson estimation, respectively. We include firm FEs (ω_i) and fiscal year FEs (μ_t) to control for time trends affecting all firms. The vector of firm controls, X_{it} , is identical to Equation (2) but varies over time.

Second, to determine whether the pre- and post-ISS treatment results in engagement variation, we also estimate Equation (3) after replacing the ISS indicator with six indicator variables (ISS^{-1} , ISS^0 , ISS^{+1} , ISS^{+2} , ISS^{+3} , and ISS^{+4}) that indicate the year relative to the ISS treatment with a low SOP vote. The values for years -4 , -3 , and -2 are included in the intercept to serve as the baseline comparison. The coefficients for ISS^{-1} and ISS^0 are important because they reveal any differences in engagement between treatment and control groups before the low SOP vote, which formally attempts to falsify the parallel trends assumption.¹⁹

The coefficient for ISS^{+1} is important because it shows how quickly firms change their shareholder engagement activities in response to ISS treatment. ISS considers the timing of shareholder engagement in its evaluation of a sufficiently robust response. The coefficients for ISS^{+2} , ISS^{+3} , and ISS^{+4} are also important because they capture longer-term differences in engagement. If firms continue to engage at a higher rate in future years, some of these coefficients should be positive and statistically significant. These coefficients test H3, which posits that engagement after ISS treatment will persist over time for the reasons detailed in Subsection 2.3.

We first estimate the effect of ISS treatment on shareholder engagement with only FEs. Column 1 of Table 9 shows that ISS treatment significantly increases shareholder engagement, as indicated by a coefficient of 0.214 and a t -statistic of 6.24 ($p < 0.001$) in tests of *engagement indicator*. The effect is also economically significant, with a 47% increase in shareholder engagement compared to the full-panel average (0.36). The results are similar using *engagement count* as the dependent variable.

When the ISS treatment indicator is replaced with six timing indicators in columns 2, 3, 5, and 6, the coefficients for ISS^{-1} and ISS^0 are not significantly different from zero. This suggests that the parallel trends assumption is not falsified.

However, the coefficient for ISS^{+1} is significantly positive at the 1% level in all four tests, indicating that the impact of ISS treatment on engagement is immediate in the first year after a low SOP vote. In column 3, the ISS treatment coefficient represents a 28.8-percentage-point increase in engagement, which is a 62% increase from the sample mean. This suggests that the effect of ISS on engagement following a low SOP vote is substantial.

The coefficients for ISS^{+2} and ISS^{+3} are positive and statistically significant for the presence of engagement, while the coefficient for ISS^{+2} is significant for

¹⁹ Figure IA-2 of the Internet Appendix plots mean values of engagement over time. Both treatment and control firms show similar patterns prior to the SOP vote near 70%, further supporting the parallel trends assumption.

Table 9
ISS treatment and shareholder engagement over time

Dependent variable	Engagement indicator			Engagement count		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>ISS treatment</i>	0.214*** (6.24)			0.719*** (4.26)		
<i>ISS treatment</i> ^{−1}		−0.038 (−1.33)	−0.033 (−1.11)		0.025 (0.19)	−0.047 (−0.26)
<i>ISS treatment</i> ⁰		−0.023 (−0.60)	−0.022 (−0.67)		0.263 (1.00)	0.250 (1.06)
<i>ISS treatment</i> ⁺¹		0.270*** (5.79)	0.288*** (5.22)		0.985*** (4.13)	0.918*** (3.44)
<i>ISS treatment</i> ⁺²		0.151** (2.90)	0.174** (3.10)		0.598** (1.98)	0.551* (1.76)
<i>ISS treatment</i> ⁺³		0.113* (2.19)	0.124** (2.30)		0.426 (1.30)	0.316 (0.91)
<i>ISS treatment</i> ⁺⁴		0.075 (1.05)	0.092 (1.30)		0.101 (0.30)	−0.064 (−0.19)
Regression	OLS	OLS	OLS	Poisson	Poisson	Poisson
Controls	No	No	Yes	No	No	Yes
Firm, year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm years	2,746	2,746	2,102	1,906	1,906	1,566
Adjusted (pseudo)- <i>R</i> ²	.522	.526	.529	.414	.424	.448

This table presents panel regressions of the relation between ISS treatment and shareholder engagement over time. Columns 1 to 3 present tests of *engagement indicator*. Columns 4 to 6 test *engagement count*. The *ISS treatment* coefficient in columns 1 and 4 represent ISS treatment times a post-treatment period indicator. In columns 2, 3, 5, and 6, the *ISS treatment*^{+/−*n*} indicator is the event-time *ISS treatment* indicator from Equation (3). All regressions include firm and year fixed effects. We cluster standard errors at the firm level in columns 1 and 4, and at the firm and year levels in all other columns. Variables are defined in Appendix C. **p* < .1; ***p* < .05; ****p* < .01 (using two-tailed tests).

engagement intensity. These results support H3, demonstrating that the strong treatment effect in *ISS*⁺¹ is still observed over time before it gradually decays.

Recent work expresses concern that d-i-d estimations with staggered treatment and two-way FEs may be biased. In our panel setting, treatment occurs in different years such that firms might serve as control firms in earlier periods and treated firms later in the sample. Thus, as a robustness test, we employ the staggered d-i-d methodology of [Borusyak, Jaravel, and Spiess \(2024\)](#). We estimate staggered treatment coefficients over time for tests of the *engagement indicator* and *engagement count* with firm and year fixed effects. We plot the coefficients from these tests along with 95% confidence intervals in Figure 6 and report the underlying results in Table IA-9 of the [Internet Appendix](#).

Figure 6 shows an increased engagement response following treatment in year *t* + 1 across both extensive and intensive margins that gradually diminishes over time. These results align with the results in Table 9 using the standard panel approach. While the staggered d-i-d approach reveals minor pretreatment trends, these are not consistently significant, are smaller than post-treatment estimates, and more importantly indicate *lower* engagement for treated firms, contrary to the observed rise in engagement post-ISS treatment.

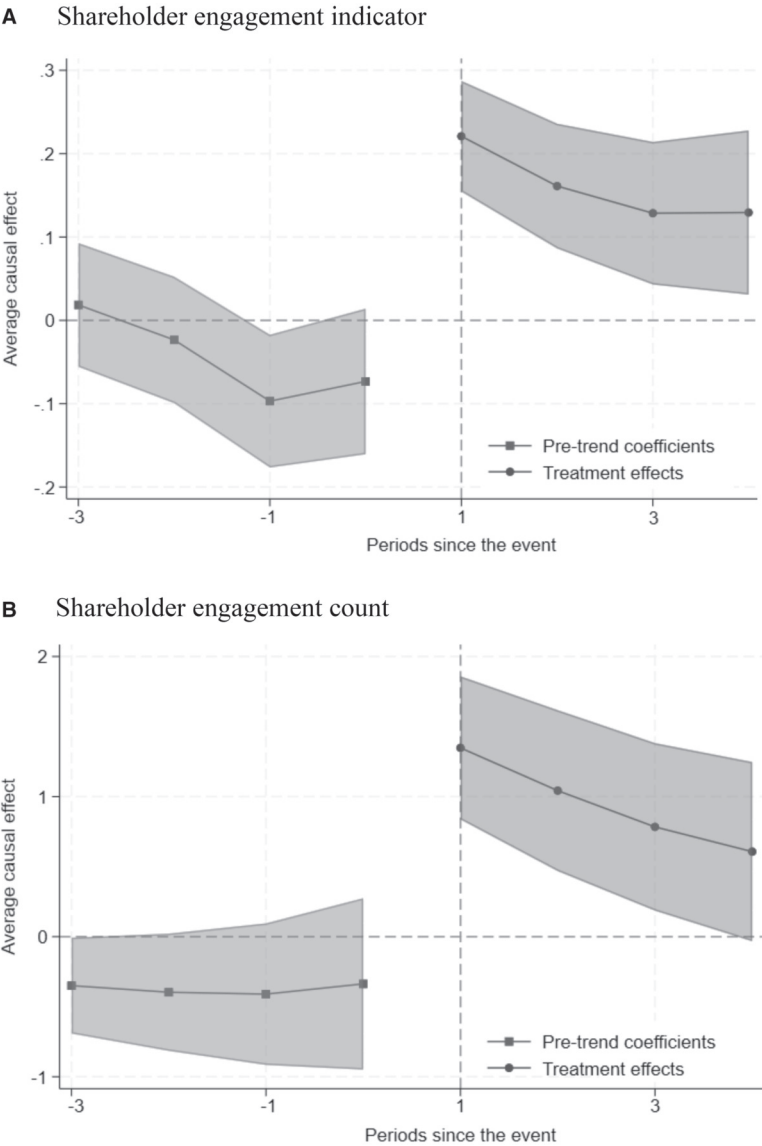


Figure 6
ISS treatment effect and shareholder engagement over time This figure displays the staggered difference-in-difference estimates for the ISS treatment effect over time, following the methodology of Borusyak, Jaravel, and Spiess (2024). ISS treated firms have SOP voting support between 67.50% and 69.99%, while control firms have SOP voting support between 70.00% and 72.50%. Squares represent the preassignment differences for ISS treated firms during the 4 years before the low SOP vote. Circles represent the estimated treatment effects for the 4 years after treatment. The shaded region depicts the 95% confidence interval for each yearly coefficient. Panel A examines the *Engagement indicator*, and panel B focuses on *Engagement count*. Detailed results are provided in Table 1A-9 of the Internet Appendix.

Notably, the magnitudes of ISS treatment on engagement are highly similar to the standard method. Specifically, the ISS treatment effect on the *engagement indicator* during the first year is 22-percentage-points higher using the staggered d-i-d methodology versus 27 percentage points in the panel regression tests in column 2 of Table 9. The *engagement count* rises by 1.35 with ISS treatment using staggered d-i-d, compared to 0.99 in the standard panel approach in column 5 of Table 9. Taken together, the results in Section 5.5 confirm the positive influence of ISS on shareholder engagement is swift, substantial, and—importantly—long-lasting, which supports H3.

4.6 Event study

Next, we use an event study to analyze the shareholder response to ISS treatment. While we recognize possible arguments for why there might be a positive, neutral, or negative shareholder response (H4a), we hypothesize the market response to ISS treatment will be positive for firms that have already received an “against” recommendation from ISS (H4b).

To test H4a and H4b, we analyze abnormal stock returns using the following framework. First, we calculate 3-day and 5-day firm-level compounded returns for the period centered on the date the firm first released the 8-K filing containing its SOP voting results.²⁰ *Abnormal returns* are calculated as the difference between the firm and CRSP value-weighted index returns during the same period.²¹ Using *abnormal returns*, we estimate OLS regressions to test the relation between ISS treatment, ISS recommendations, and shareholder returns using the following equation for H4a:

$$\begin{aligned} \text{Abnormal returns}_{it} = & \alpha + \beta_1 \text{ISS treatment}_{it} + \beta_2 \text{ISS “against” SOP}_{it} \\ & + \beta_3 \text{SOP voting support}_{it} + \phi_j + \mu_t + \varepsilon_{it}, \end{aligned} \quad (4)$$

and the following equation for H4b:

$$\begin{aligned} \text{Abnormal returns}_{it} = & \alpha + \beta_1 \text{ISS treatment}_{it} \times \text{ISS “against” SOP}_{it} \\ & + \beta_2 \text{ISS treatment}_{it} + \beta_3 \text{ISS “against” SOP}_{it} \\ & + \beta_4 \text{SOP voting support}_{it} + \phi_j + \mu_t + \varepsilon_{it}, \end{aligned} \quad (5)$$

where *ISS treatment* is set to one for firms that received between 67.50% and 69.99% SOP voting support and zero for firms that received 70.00% to 72.50%. The variable *ISS “against” SOP* is set to one if ISS recommended voting against the SOP proposal. *SOP voting support* is the disclosed percentage of

²⁰ Firms must submit the results of the SOP vote within 4 business days of the meeting, typically via 8-K Item 5.07. See <https://www.sec.gov/files/form8-k.pdf>. We download 8-K dates from WRDS SEC Analytics Suite and hand-check them for accuracy.

²¹ To omit potentially distressed firms, we drop those with a stock price below \$1 per share.

Table 10
Abnormal stock returns around SOP vote disclosure

	(1)	(2)	(3)	(4)	(5)	(6)
	All SOP vote results				Removing contaminated disclosures	
Abnormal returns period:	3-day [−1,+1]	5-day [−2,+2]	3-day [−1,+1]	5-day [−2,+2]	3-day [−1,+1]	5-day [−2,+2]
<i>ISS treatment</i>	0.004 (0.38)	−0.000 (−0.00)	−0.008 (−0.63)	−0.017 (−1.17)	−0.009 (−0.74)	−0.015 (−1.09)
<i>ISS “against” SOP</i>	0.010* (1.80)	0.008 (1.32)	0.002 (0.20)	−0.004 (−0.41)	−0.001 (−0.10)	−0.002 (−0.26)
<i>SOP voting support</i>	0.266 (0.73)	0.143 (0.35)	0.279 (0.77)	0.161 (0.39)	0.428 (1.18)	0.292 (0.73)
<i>ISS treatment × ISS “against” SOP</i>			0.019* (1.72)	0.026** (2.11)	0.021* (1.95)	0.023* (1.89)
Regression	OLS	OLS	OLS	OLS	OLS	OLS
Year, industry FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	399	399	399	399	371	371
Adjusted R^2	−.016	−.016	−.010	−.007	−.005	−.004

This table reports tests of abnormal returns around the disclosure of say-on-pay (SOP) voting results for the sample. We compute firm-level stock returns by calculating the compound daily returns during the 3-day and 5-day period centered on the date the firm first releases the SOP voting results, typically in an SEC 8-K filing (Item 5.07). We then estimate *abnormal returns* by subtracting the compound returns on the CRSP value-weighted index return with distributions during the same period. Using ordinary least squares (OLS) regressions, columns 1 through 4 include tests of abnormal returns for the full sample of ISS treatment and control firms with available return information and a stock price greater than \$1.00 on the date before voting disclosure. Columns 5 and 6 remove “contaminated” 8-K filings that contain other information items, such as material contracts or earnings information. *ISS treatment* equals one for firms that receive between 67.50% and 69.99% SOP voting support, and zero for firms that receive 70.00% to 72.50%. *ISS “against” SOP* equals one if ISS recommends voting against SOP, and otherwise zero. The variable of interest is the interaction term, *ISS treatment × ISS “against” SOP*, which equals one for firms that are both treated with a below 70% vote and receive an ISS recommendation to vote against SOP prior to the meeting, and otherwise zero. *SOP voting support* is the disclosed percentage of shareholder votes that support the SOP proposal divided by the number of votes for and against SOP. All regressions include year and industry fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

shareholder votes that supported the SOP proposal. By examining a narrow bandwidth around 70% and controlling for SOP voting support as well as the ISS treatment effect, this specification is functionally similar to a regression discontinuity design. We include industry FEs (ϕ_j) using one-digit SIC codes to control for time-invariant unobserved heterogeneity across industries, and fiscal year FEs (μ_t) to control for time trends that affect all firms. We present the results of this regression in Table 10.

For H4a, the variable of interest is *ISS treatment*, and results are provided in columns 1 and 2. The coefficients for *ISS treatment* for both the 3-day and 5-day regressions are insignificant, failing to reject the null hypothesis in H4a that there is no shareholder response for treated firms. This could be due to the smaller sample size, or perhaps it is simply the case that investors do not anticipate treated firms, on average, to subsequently take engagement actions that will influence shareholder value.

Next, we test H4b, which is that treated firms that had already received an against recommendation from ISS will see a positive stock return due to the threat being more credible. The variable of interest in this analysis is the

interaction term $ISS\ treatment \times ISS\ "against" SOP$, which is set to one for firms that are both treated with a vote below 70% and receive an ISS recommendation to vote against SOP before the meeting, and otherwise zero. Results testing this hypothesis are reported in columns 3 and 4 of Table 10.

In each of these columns, the coefficients for the interaction term $ISS\ treatment \times ISS\ "against" SOP$ is positive and statistically significant at the 10% level or higher for both the 3- and 5-day windows. This suggests that shareholders view strengthened engagement as beneficial when coupled with a clear threat of economic sanctions by ISS. The effect is also economically meaningful, as the results equate to positive abnormal returns between about 2% and 3%.

In columns 5 and 6, we check the robustness of these findings by excluding 28 observations from our sample with potentially contaminated information that could influence the abnormal returns. Specifically, we drop observations in which the 8-K also includes material agreements or financial results (8-K Items 1.01 and 2.02). The regressions yield similar results to those reported in columns 3 and 4. In Table IA-10 of the Internet Appendix, we find similar results using an RD estimation. In aggregate, the results show that the market perceives ISS-induced engagement to be value-adding when coupled with a greater threat of future sanctions.

5. Additional Tests

5.1 Role of agency conflicts in engagement

Our evidence that shareholder engagement increases following ISS intervention, and that shareholders appear to value it when ISS's threat is strongest, raises the question of why these firms did not proactively communicate or invest in necessary engagement infrastructure before ISS's involvement. The answer could simply be that once firms started engaging, the marginal benefits of continued engagement exceeded the marginal costs, as outlined in H3. However, it might also be related to deficiencies in firm governance, which would imply that ISS plays a broader governance role for these firms.

In this section, we conduct preliminary analyses to examine whether director agency costs associated with engagement (i.e., directors may prefer a "quiet life") might be an explanation for why some firms engaged less prior to ISS treatment. By not engaging with shareholders, entrenched boards could avoid the discomfort of addressing shareholder concerns about executive compensation. We test this possibility by examining if boards with properties of higher agency conflicts are likely to increase their engagement more following ISS intervention.

We use two measures as proxies for board-level agency conflicts. Prior work shows that long-serving directors on the compensation committee tend to be more management-friendly, which correlates with higher CEO pay (Vafeas 2003). Thus, for our first measure, we use *CC tenure*, which is the average

Table 11
Shareholder engagement and agency conflicts

	(1)	(2)	(3)	(4)
	Engagement indicator		Engagement count	
	CC tenure	Classified boards	CC tenure	Classified boards
ISS treatment	−0.059 (−0.64)	0.107* (1.89)	−0.025 (−0.09)	0.223 (1.58)
Agency variable	−0.011* (−1.87)	−0.101 (−1.63)	−0.042** (−2.28)	−0.344* (−1.68)
ISS treatment × agency variable	0.025*** (2.68)	0.123 (1.42)	0.050** (2.02)	0.507* (1.94)
Regression	OLS	OLS	Poisson	Poisson
Controls	Yes	Yes	Yes	Yes
Year, industry FE	Yes	Yes	Yes	Yes
Observations	426	426	426	426
Adjusted R ²	.262	.253	.230	.231

This table presents regressions of the heterogeneous effect of ISS treatment on shareholder engagement in year $t+1$ based on measures of board agency conflicts. The dependent variables are *engagement indicator*, which we test using OLS regressions, and *engagement count*, which we test using Poisson regressions. We assess heterogeneous responses to ISS treatment across firms using two ex ante board agency variables: *CC tenure* and *Classified boards*. The focal variable in each analysis is *ISS treatment*, which is interacted with these board agency variables to determine whether the ISS treatment effect differs for firms with higher baseline measures of agency conflicts. All regressions include year and industry fixed effects. We define variables in Appendix C. * $p < .1$; ** $p < .05$; *** $p < .01$ (using two-tailed tests).

board service length in years for compensation committee members. Our second measure is the presence of *classified boards*. Several studies link board classification to managerial entrenchment (e.g., [Faleye 2007](#); [Bebchuk, Cohen, and Ferrell 2009](#)), which suggests that the entire management team in such firms would prefer lesser interactions with investors.²²

Next, we adjust Equation (2) to include *CC tenure* and *classified boards* and their interactions with ISS treatment, with the dependent variables being *engagement indicator* and *engagement count*. If firms whose directors have higher agency costs intensify their engagement efforts at a greater rate, then we would expect a positive coefficient for the interaction of *ISS treatment* × *CC tenure* and *ISS treatment* × *classified boards*.

Table 11 shows that the effect of ISS treatment on the extensive and intensive margins of engagement is indeed higher for boards with greater *CC tenure*. Specifically, the coefficient for *ISS treatment* × *CC tenure* is positive and significant at the 5% level or better in columns 1 and 3. For firms with classified boards, we find that the intensive margins of engagement increase at a significantly greater rate following ISS intervention. While the coefficient for the extensive margin is positive, it falls short of significance. Collectively,

²² ISS considers classified boards as a “problematic governance structure” in its proxy voting guidelines. In a takeover context, some studies find that classified boards do not destroy value (e.g., [Bates, Becher, and Lemmon 2008](#); [Dey and White 2021](#)).

Table 12
Threshold analysis

	(1)	(2)	(3)	(4)	(5)
	Univariate		RD		N
Threshold (%)	Placebo treated difference	<i>p</i> -value	Placebo treated difference	<i>p</i> -value	[PT:PC]
50	−0.034	.686	0.112	.743	50:89
55	0.003	.968	0.210	.388	110:101
60	−0.032	.573	−0.228	.156	127:150
65	−0.040	.425	−0.044	.774	175:217
70 (ISS)	0.149***	.002	0.334***	.010	209:217
75	0.066	.107	−0.233*	.066	272:312
80	−0.017	.621	−0.106	.349	398:412
85	0.005	.864	0.133	.179	515:617
90	0.035	.114	−0.032	.642	732:1168

This table conducts placebo tests to evaluate the probability of engagement in year $t+1$ for firms within $\pm 2.5\%$ around each say-on-pay (SOP) voting support threshold between 50% and 90%. Column 1 presents the placebo treated difference in the *engagement indicator* between firms just below (placebo treated) each threshold versus those just above it (control). We test that this difference is significantly different from zero using two-tailed t -tests and report the corresponding p -value in column 2. Columns 3 and 4 present information from regression discontinuity (RD) tests of the placebo treated difference in the *engagement indicator* for firms just below each threshold versus those just above it. We report the robust bias-corrected coefficient for placebo treatment in column 3 using local linear regression discontinuity estimations from Cattaneo, Jansson, and Ma (2018). The corresponding p -value is reported in column 4. Column 5 presents the number of placebo treated firms (PT) and control firms (PC). For comparison, we report the ISS treated values with the gray-shaded line for the 70% threshold, which correspond to values in Tables 5 and 6 for ISS treated versus control firms. * $p < .1$; ** $p < .05$; *** $p < .01$.

the results in Table 11 suggest that firms with weaker ex ante board oversight tend to engage more with shareholders after receiving ISS treatment, pointing to a channel through which ISS plays a positive governance role in capital markets.

5.2 Placebo tests

Our engagement analyses rely on the 70% SOP voting support threshold as being an important threshold based on ISS’s policies. In our final analysis, we conduct placebo tests to assess if firms simply increase engagement whenever they fall below a salient threshold. For these tests, we rerun our univariate and RD analyses of the extensive margins of engagement for a bandwidth of $\pm 2.5\%$ around nine thresholds from 50% to 90%, in 5% increments, and report the results in Table 12. We conduct additional RD placebo analyses in Table IA-11 and Figure IA-3 of the Internet Appendix.

The univariate tests in column 1 of Table 12 show no statistically significant differences in the extensive margin of engagement except at the 70% threshold. None of the other thresholds from 50% to 90% are significantly different from zero in univariate tests. Using the RD approach, we find marginally significant differences at the 75% threshold. However, the sign of the coefficient for falling just below 75% is negative, suggesting that engagement is actually lower just below that threshold, the opposite of what we would expect if this threshold was important. None of the other thresholds except ISS’s 70% threshold

exhibit significant differences in engagement using the RD approach. These findings reinforce our conclusion that ISS encourages shareholder engagement following a low SOP vote (in support of H1), and that the engagement outcomes are not simply a reflection of a salient threshold.

6. Conclusion

Our study presents novel evidence on how ISS shapes firms' information and governance environments through their influence on shareholder engagement. We exploit a quasi-natural experiment by studying firms that receive a level of SOP voting dissent above which ISS threatens to impose economic sanctions if managers and directors do not engage with shareholders and address concerns. SOP voting outcomes are smoothly distributed across ISS's policy threshold, with firms near the threshold showing similar *ex ante* characteristics. This suggests locally random exposure to ISS treatment.

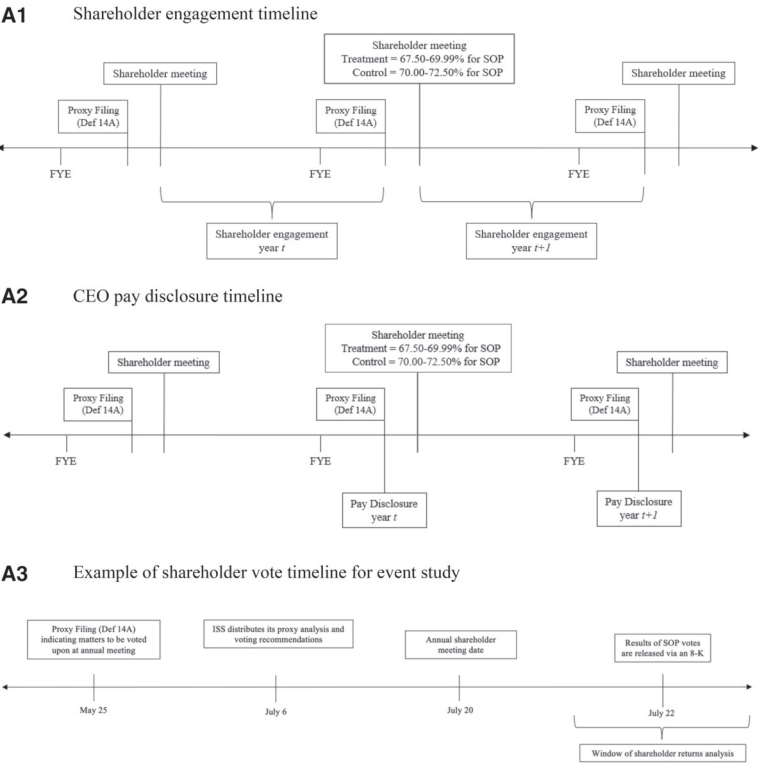
Utilizing average treatment effect and regression discontinuity methodologies, we find that ISS treatment leads to a significant increase in the extensive and intensive margins of shareholder engagement in the year after a low SOP vote. This increase is higher when the threat of ISS sanctions against directors' board reelections is greater. Difference-in-differences tests show the increased engagement persists after ISS stops enhanced monitoring of the firm, implying that the initial ISS scrutiny has a lasting impact on the firm's engagement policies with shareholders. Event study analyses indicate that these interactions with shareholders appear to be value-enhancing, especially when combined with greater ISS scrutiny.

Our study enhances the literature on the role of ISS in capital markets, specifically its effect on information flows between firms and investors (e.g., Ertimur, Ferri, and Oesch 2013; McCahery, Sautner, and Starks 2016; Hayne and Vance 2019). We establish that ISS facilitates ongoing, multifaceted communication, which can enable firms to better incorporate shareholder preferences for corporate strategies. Such proactive, bidirectional engagement likely improves the quality of information exchange beyond traditional one-directional voting outcomes (e.g., Kakhbod et al. 2023).

A better understanding of the role and influence of proxy advisors, especially that of ISS, is crucial to guiding policy debates on the scope and influence of ISS on corporate governance and shareholder voting. Our results provide causal evidence on ISS's influence on firms' shareholder engagement behaviors and allow for more informed deliberations on the regulatory concern that ISS promotes a one-size-fits-all approach to governance. Our study provides a more nuanced understanding of the governance role of proxy advisors in capital markets and raises interesting questions for future research, such as exploring optimal levels of shareholder engagement in firms.

Code Availability: The replication code is available in the Harvard Dataverse at <https://doi.org/10.7910/DVN/ABLKE4>.

Appendix A. Timeline of Events



Appendix B. Examples of Shareholder Engagement Disclosure

Table B1
Seacoast Banking Corporation of Florida's 2017 proxy statement

Shareholder Engagement and Board Responsiveness	
The company engages with our shareholders to ensure that the Board and management are aware of and address issues of importance to our investors. We regularly meet with various institutional shareholders and welcome feedback from other shareholders, which is considered by the Board or appropriate Board committees.	
The company communicates with shareholders in a number of ways: • Discussing important business developments and answering shareholder questions at annual meetings; • Holding conference calls regarding quarterly earnings and significant corporate developments; • Through this proxy statement, chairman's letters, news releases, website, participation in industry conferences, and other periodic events.	
The Company's Corporate Governance Guidelines provide for a process by which shareholders may communicate with the Board, a Board committee or the non-management directors as a group, or other individual directors. Shareholders who wish to communicate with the Board of Directors, a Board committee, the Lead Independent Director, other directors or an individual director may do so by sending written communications addressed to the Board of Directors, a Board committee or such group of directors or individual director, c/o Corporate Secretary, Seacoast Banking Corporation of Florida, 811 Colorado Avenue, P.O. Box, 9012, Stuart, Florida 34995. All communications will be compiled by the Company's Secretary and submitted to the Board of Directors, a committee of the Board of Directors or the appropriate group of directors or individual director, as appropriate, at the next regular meeting of the Board.	
Since 2009 the Company has annually included in its proxy statement a separate advisory vote on the compensation paid to its executives, as disclosed in the Compensation Discussion and Analysis, the compensation tables and related proxy disclosure, commonly known as a "say-on-pay" proposal. Independent surveys have shown that an annual vote is the preferred frequency of most institutional investors. Our Board also endorses an annual vote as we believe it gives shareholders an opportunity to voice their concerns with respect to executive compensation. Shareholder support of our say-on-pay proposal at our 2016 annual meeting declined compared to prior years. (See "Outcome of our 2016 Say-On-Pay vote" in the table below.) Shareholder support of directors standing for re-election at the 2016 annual meeting also dropped compared to prior year. To better understand shareholder interests and concerns, we expanded our shareholder outreach in 2016, taking the following actions: • Members of our Board and executive management team met in-person and by phone with shareholders representing a significant portion of Seacoast's outstanding unaffiliated shares. • In March 2016, the Company entered into an Observer Rights Agreement with one of our largest shareholders, Bearwood Capital Management, L.L.C. ("Bearwood") and Matthew Lindenhum, a principal of Bearwood, pursuant to which we gave Mr. Lindenhum the right to attend all meetings of our Board of Directors in a non-voting observer capacity, and to receive materials provided to board members, subject to certain exceptions. The agreement can be terminated by either party at any time, but remains in effect. Mr. Lindenhum's viewpoint on matters impacting institutional investors has been very insightful and beneficial to the Company; • We have hosted visits with a number of large investors at our facilities in Stuart to provide them with a better understanding of the depth and talent of our management team, the execution of our balanced strategy and our progress in creating shareholder value. • We hosted an Investor Day and webcast on February 22, 2017, where Seacoast's management team provided a detailed update of the Company's strategy and long-term vision.	
In these meetings, our shareholders expressed a wide range of viewpoints relating to our performance, compensation, governance and other matters. This engagement process was very informative and constructive. Below are highlights of the feedback we have received from shareholders and our Board's response:	
What We Heard	Our Board's Response
Improve financial performance to deliver results expected from acquisitions	<i>Delivered Promised Results:</i> Achieved our 2016 earnings target of \$1.00 fully diluted adjusted earnings per share ¹ ("EPS") despite economic headwinds. Improved adjusted efficiency ratio ¹ by 12% year over year, and from 74.9% in fourth quarter of 2014 to 60.8% in the fourth quarter of 2016.
Higher stock ownership by management and directors	<i>Replacing Cash Bonuses with Equity:</i> Replaced 2016 cash bonuses paid to executive officers for achievement of performance objectives with performance based and performance-contingent stock awards. All of our directors are paid a stock retainer; several participated in our recent capital raise.
Reduce Board tenure and the risk of entrenchment	<i>Three New Directors:</i> In 2016, our Board appointed two new directors, Timothy S. Hural and Herbert Luis, further enforcing its commitment to a balanced mix of new directors with fresh perspectives and, for continuity, seasoned, experienced directors with deep knowledge of the Company and its markets. In 2017, our Board has nominated another new Board member, Alvarez Monserrat, to replace a longer-tenured director. In addition, two long-tenured directors rotated off the COC in 2016 and one short-tenured director was added, resulting in the majority of the COC now comprised of short-tenured directors.

¹ Non-GAAP measure; refer to Appendix A - Information Regarding Non-GAAP Financial Measures.

What We Heard	Our Board's Response
Concerns regarding our severance payments under Change-in-Control agreements and award agreements	<i>Modifications to Existing Named Executive Officer ("NEO") Change-in-Control ("CIC") Agreements:</i> In September 2016, we entered into new CIC agreements with our NEOs which: i) eliminate excise tax gross-up payments, and ii) do not provide benefits that are paid before a CIC closes or if the acquirer retains the executive. Since 2015, our equity award agreements require a CIC and job loss or failure of the acquirer to convert awards into new issue awards before vesting occurs.
Outcome of our 2016 Say-On-Pay vote	At our 2016 annual meeting of shareholders, our say-on-pay proposal received the support of 67.6% of the votes cast. In contrast, since 2009, our previous annual say-on-pay proposals received support of over 94% of the votes cast. Our COG considered the vote in relation to: 1) the alignment of our compensation program with the long-term interests of our shareholders, 2) the evolution of our business strategy with emerging opportunities and in fulfilling customer demand for innovative products and services, and 3) the relationship between risk-taking and the incentive compensation provided to our executives. The COG will continue to evaluate and refine our executive compensation plan to place greater emphasis on creating shareholder value, long-term performance and profitability based on emerging opportunities.
Suggestion to better articulate our business strategy	We have redoubled our efforts in our quarterly communications with shareholders and investors, and our enhanced our investor outreach, to convey our balanced business strategy of: 1) organic growth, 2) accretive and thoughtful M&A, and 3) innovation and digitally-engaged distribution. We formed the S&I Committee to provide oversight of these efforts.

The Board and the COG will continue to monitor best practices, future advisory votes on executive compensation and other shareholder feedback to guide it in addressing issues of importance to our investors.

Source: https://www.sec.gov/Archives/edgar/data/730708/000114420417019335/v463271_def14a.htm

Table B2
Ventas, Inc.'s 2017 proxy statement

Responsive Redesign Following 2016 Advisory Vote on Executive Compensation and Stockholder Outreach

We submit an advisory vote to our stockholders on an annual basis to approve our executive compensation. At our 2016 Annual Meeting of Stockholders, holders of 68% of the shares represented at the meeting voted to approve, on an advisory basis, our executive compensation. Although a significant majority continued to support our executive compensation program, the result of the 2016 vote was in contrast to the strong support we received in prior years, specifically 95% in 2015 and 90% in 2014. The level of support declined despite (a) the absence of material changes to our general executive compensation program between the 2015 and 2016 Annual Meetings, (b) in the view of our Board, the consistent strong alignment between our executive pay and performance (see "Strong Pay-for-Performance Alignment" chart above) and (c) the significant reduction in long-term equity incentive awards granted to our Named Executive Officers in early 2016 resulting from our bottom quartile TSR performance as measured against our peer group for the one- and three-year measurement periods ending December 31, 2015, despite our outstanding long-term TSR outperformance.

We have continued to gain valuable insight from engaging with our stockholders on a consistent basis. We have conducted broad investor outreach programs on three separate occasions. In late 2015 and early 2016, the Chair of our Compensation Committee, our independent compensation consultant to the Compensation Committee and members of our Legal team reached out to 27 of our largest stockholders (holding more than 60% of our outstanding shares of common stock) to discuss our executive compensation program and invited such stockholders to provide us with feedback on our executive compensation program and corporate governance practices.

After filing our 2016 Proxy Statement, the Chair of our Compensation Committee, our independent compensation consultant to the Compensation Committee and members of our Legal team again reached out to our 30 largest stockholders (holding more than 60% of our outstanding shares of common stock) to discuss, and solicit feedback regarding potential changes to, our executive compensation program. We invited such stockholders to provide us with constructive comments regarding our executive compensation program and corporate governance practices.

In early 2017, a combination of the Chair of our Compensation Committee and members of our Legal team again reached out to our 32 largest stockholders (holding 60% of our outstanding shares of common stock) to discuss, and solicit feedback regarding our executive compensation program. We invited such stockholders to provide us with constructive comments regarding our executive compensation program and corporate governance practices.

Based on these discussions, we learned that our stockholders:

- ✓ generally approve of the overall structure of our executive compensation program and diversity of goals, particularly our use of balanced metrics of growth, risk management and capital structure to mitigate risk and promote responsible, sustained long-term growth;
- ✓ generally approve of our implementation of the executive compensation program, the factors considered and the decisions made under the program;
- ✓ generally approve of our proxy disclosures regarding our executive compensation program and corporate governance best practices;
- ✓ generally support our pay-for-performance alignment; and
- ✓ generally endorse our corporate governance practices.

We also received constructive feedback from our investors. After careful consideration of this feedback, we have decided to make significant and responsive modifications to our executive compensation program, beginning with the 2017 compensation cycle.

What We Heard From Investors

Changes We Made

Move to a forward-looking equity incentive plan

Reduce the percentage of discretionary in long-term equity incentive plan

Extend the measurement period to the long-term equity incentive plan

Eliminate single-trigger Change in Control vesting

Consider proxy access

Performance-based awards are now based on future performance

Eliminated qualitative or discretionary goals previously comprising 50% of award opportunities

More balanced mix of performance metrics

25% TSR relative to the S&P 500 index

3-year

Proxy access

Increased performance-based component

60% performance-based restricted stock units (RSUs)

40% new-based restricted stock units (RSUs)

17.5% TSR relative to the FTSE 1000 index

3-year

Estimated immediate vesting and moved equity vesting period from 1 year to 3-year

The key features of the new program are consistent with the feedback we have received from our largest stockholders and are described in detail below in the section entitled, "Going Forward: Responsive Redesign of Long-Term Incentive Compensation."

We believe this redesigned program enhances alignment of pay and performance, is responsive to investor feedback and provides simpler, clear objectives, while achieving our goals of attracting, retaining and motivating talented executives and rewarding superior performance in the context of our business risk environment.

Source: https://www.sec.gov/Archives/edgar/data/740260/000110465917021407/a17-2391_1def14a.htm#ResponsiveRedesignFollowing2016A_120349

Appendix C. Variable Definitions

Table C1
Variable definitions

Variable	Definition
Engagement	
Engagement indicator	Equals one if the proxy statement mentions any of the engagement keywords engage*, feedback*, and conversation* within 100 characters on either side of the shareholder keywords shareholder* or stockholder*
Engagement count	A count of the number of engagement keywords near shareholder keywords in the proxy statement
Engagement references SOP	Equals one if the shareholder engagement discussion in the proxy statement references the say-on-pay (SOP) vote from the prior year, and otherwise zero
Engagement table	Equals one if the firm provides a table of investor concerns and the board response following engagement in the proxy statement, and otherwise zero
Shareholders contacted	Percentage of shares outstanding held by investors that were contacted during engagement, from the proxy statement
Shareholders spoken with	Percentage of shares held by investors spoken with during engagement, from the proxy statement

Table C1
(Continued)

Variable	Definition
<i>Firm characteristics</i>	
<i>SOP voting support</i>	Votes for the SOP proposal divided by the sum of the votes for and against SOP; Abstention votes are not included based on ISS's definition
<i>SOP voting dissent</i>	Votes against the SOP proposal divided by the sum of votes for and against
<i>ISS "against" SOP</i>	Equals one if ISS recommends voting against SOP and otherwise zero
<i>GL "against" SOP</i>	Equals one if Glass Lewis recommends voting against SOP and otherwise zero
<i>Size</i>	Natural logarithm of total assets
<i>Market-to-book</i>	Market value of equity plus debt divided by total assets
<i>Return-on-assets</i>	Earnings before interest, taxes, depreciation, and amortization over assets
<i>Leverage</i>	Long-term debt plus long-term debt in current liabilities divided by total assets
<i>Sales growth</i>	Year-over-year growth in total revenue
<i>Abnormal returns</i>	Annual common stock return less the return of the value weighted CRSP index during the fiscal year
<i>Stock return volatility</i>	Standard deviation of the monthly stock returns during the fiscal year
<i>Operating loss</i>	Equals one if the firm has a nonmissing value of tax loss carry-forward in Compustat during the fiscal year, and otherwise zero
<i>CD&A exempt</i>	Equals one if the firm meets the SEC's definition of a Smaller Reporting Company and is exempt from the CD&A disclosure, and otherwise zero
<i>ln(analysts)</i>	Log transformation of one plus the number of analysts providing earnings per share forecasts during the four quarters prior to the annual meeting, from IBES
<i>Institutional ownership (IO)</i>	Percentage of shares held by institutions with more than \$100 million in assets under management as reported on SEC Form 13-F, from Thomson
<i>Quasi-indexer IO</i>	Percentage of shares held by institutions with a long-term horizon, low portfolio turnover, and greater diversification based on Bushee (1998)
<i>Transient IO</i>	Percentage of shares held by institutions with a short-term horizon, high portfolio turnover, and greater diversification based on Bushee (1998)
<i>Dedicated IO</i>	Percentage of shares held by institutions with a long-term horizon, low portfolio turnover, and less diversification based on Bushee (1998)
<i>CEO turnover</i>	Equals one for firms with any change in CEO based on hand-collected proxy filing information, and otherwise zero
<i>Adjusted stock returns</i>	The firm's 12-month stock return minus its two-digit SIC industry's market-cap-weighted average return
<i>Adjusted ROA</i>	The firm's income before extraordinary items as a percentage of assets minus the two-digit SIC industry median for the same measure
<i>Net income</i>	Income or loss post-expenses, including extraordinary items and discontinued operations using Compustat item "NI"
<i>CEO pay</i>	
<i>Equity pay</i>	Sum of CEO stock and option awards from the Summary Compensation Table (SCT)
<i>Nonequity pay</i>	Sum of CEO salary, bonus, nonequity incentives, change in pension value, and other compensation from the SCT
<i>Expected pay</i>	Estimated CEO total compensation using OLS regression based on Rouen (2020), as well as prior stock returns, stock return volatility, book-to-market ratio, book leverage, and indicator for firms with inside hires as well as negative net income
<i>Unexpected pay</i>	The reported CEO compensation in the SCT less expected pay
<i>Governance or directors</i>	
<i>CC tenure</i>	Average board service length (in years) for compensation committee (CC) members
<i>Classified boards</i>	Equals one if the firm's board is divided into separate classes with each class being elected to overlapping terms (also known as staggered boards) and otherwise zero
<i>Board independence</i>	The ratio of the number of independent directors to the total number of directors
<i>Dual class shares</i>	Equals one if the firm has a structure where more than one class of shares are outstanding, with each offering different voting rights and otherwise zero
<i>CC member voting support</i>	Average percentage of votes for directors on the compensation committee divided by the sum of the votes for and against members of the compensation committee
<i>CC chair voting support</i>	Percentage of votes for the chair of the compensation committee divided by the sum of the votes for and against the chair of the compensation committee

Table C1
(Continued)

Variable	Definition
<i>Director voting support</i>	Average percentage of votes for all directors divided by the sum of the votes for and against members of the board
<i>Directors voted on</i>	The total count of directors subject to shareholder voting at the next annual meeting
<i>CC members voted on</i>	The total count of compensation committee members subject to shareholder voting at the next annual meeting
<i>CC chair voted on</i>	Equals one if the chair of the compensation committee is subject to shareholder voting at the next annual meeting and otherwise zero
<i>Less responsive</i>	Equals one if the firm does not disclose shareholder engagement (i.e., <i>engagement indicator</i> is equal to zero) and otherwise zero
<i>More responsive</i>	Equals one if the firm discloses shareholder engagement (i.e., <i>engagement indicator</i> is equal to one) and otherwise zero

References

- Albuquerque, A., M. E. Carter, and S. Gallani. 2020. Are ISS recommendations informative? Evidence from assessments of compensation practices. Working Paper, HBS.
- Alexander, C., M. Chen, D. Seppi, and C. Spatt. 2010. Interim news and the role of proxy voting advice. *Review of Financial Studies* 23:4419–54.
- Bach, L., and D. Metzger. 2019. How close are close shareholder votes? *Review of Financial Studies* 32:183–214.
- Balsam, S., J. Boone, H. Liu, and J. Yin. 2016. The impact of say-on-pay on executive compensation. *Journal of Accounting and Public Policy* 35:162–91.
- Bates, T. W., D. A. Becher, and M. L. Lemmon. 2008. Board classification and managerial entrenchment: Evidence from the market for corporate control. *Journal of Financial Economics* 87:656–77.
- Bebchuk, L., A. Cohen, and A. Ferrell. 2009. What matters in corporate governance? *Review of Financial Studies*, 22:783–827.
- Berg, T., M. Reisinger, and D. Streitz. 2021. Spillover effects in empirical corporate finance. *Journal of Financial Economics* 142:1109–27.
- Boone, A. L., and J. T. White. 2015. The effect of institutional ownership on firm transparency and information production. *Journal of Financial Economics* 117:508–33.
- Borusyak, K., X. Jaravel, and J. Spiess. 2024. Revisiting event study designs: Robust and efficient estimation. *Review of Economic Studies*. Advance Access published February 6, 2024, 10.1093/restud/rdae007.
- Bushee, B. 1998. The influence of institutional investors on myopic R&D investment behavior. *Accounting Review* 73:305–33.
- Cabazon, F. 2024. Executive compensation: The trend toward one size fits all. *Journal of Accounting and Economics*. Advance Access published July 3, 2024, 10.1016/j.jacceco.2024.101708.
- Cai, J., J. L. Garner, and R. A. Walkling. 2009. Electing directors. *Journal of Finance* 64:2389–421.
- Calonico, S., M. Cattaneo, and M. Farrell. 2020. Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *Econometrics Journal* 23:192–210.
- Calonico, S., M. Cattaneo, M. Farrell, and R. Titiunik. 2017. rdrobust: Software for regression-discontinuity designs. *Stata Journal* 17:372–404.
- Cattaneo, M., M. Jansson, and X. Ma. 2018. Manipulation testing based on density discontinuity. *Stata Journal* 18:234–61.
- . 2020. Simple local polynomial density estimators. *Journal of the American Statistical Association* 115:1449–55.

- Cattaneo, M., and R. Titiunik. 2022. Regression discontinuity designs. *Annual Review of Economics* 14:821–51.
- Choi, S., J. Fisch, and M. Kahan. 2009. The power of proxy advisors: Myth or reality. *Emory Law Journal* 59:869–918.
- Clayton, J. 2018. Statement Announcing SEC Staff Roundtable on the Proxy Process. Speech, July 30. <https://www.sec.gov/newsroom/speeches-statements/statement-announcing-sec-staff-roundtable-proxy-process>
- Cohn, J. B., Z. Liu, and M. I. Wardlaw. 2022. Count (and count-like) data in finance. *Journal of Financial Economics* 146:529–51.
- Copland, J., D. Larcker, and B. Tayan. 2018. The big thumb on the scale: An overview of the proxy advisory industry. Rock Center for Corporate Governance at Stanford University, CGRP-72, 18–27.
- Crawford, V., and J. Sobel. 1982. Strategic information transmission. *Econometrica* 50:1431–51.
- Del Guercio, D., L. Seery, and T. Woidtke. 2008. Do boards pay attention when institutional activists “just vote no”? *Journal of Financial Economics* 90:84–103.
- Delmas, M., and M. Toffel. 2008. Organizational responses to environmental demands: Opening the black box. *Strategic Management Journal* 29:1027–55.
- Denis, D. K., T. Jochem, and A. Rajamani. 2020. Shareholder governance and CEO compensation: The peer effects of say on pay. *Review of Financial Studies* 33:3130–73.
- Dey, A., and J. T. White. 2021. Labor mobility and antitakeover provisions. *Journal of Accounting and Economics* 71:101388.
- Dong, Y., O. Li, Y. Lin, and C. Ni. 2016. Does information-processing cost affect firm-specific information acquisition? Evidence from XBRL adoption. *Journal of Financial and Quantitative Analysis* 51:435–62.
- Ellickson, R. 1991. *Order without law*. Cambridge, MA: Harvard University Press.
- Ertimur, Y., F. Ferri, and D. Oesch. 2013. Shareholder votes and proxy advisors: Evidence from say on pay. *Journal of Accounting Research* 51:951–96.
- Fairfax, L. 2013. Mandating board-shareholder engagement. *University of Illinois Law Review* 2013: 821–58.
- Faleye, O. 2007. Classified boards, firm value, and managerial entrenchment. *Journal of Financial Economics* 83:501–29.
- Fama, E., and M. Jensen. 1983. Separation of ownership and control. *Journal of Law and Economics* 26: 301–25.
- Fich E., and A. Shivdasani. 2007. Financial fraud, director reputation, and shareholder wealth. *Journal of Financial Economics* 86:306–36.
- Financial Economists Roundtable. 2022. The controversy over proxy voting: The role of asset managers and proxy advisors. https://www.financialeconomistsroundtable.com/s/FER-2022-Statement-on-Proxy-Voting_final.pdf
- Gelman, A., and G. Imbens. 2019. Why high-order polynomials should not be used in regression discontinuity designs. *Journal of Business & Economic Statistics* 37:447–56.
- Gerling, K., H. P. Grüner, A. Kiel, and E. Schulte. 2005. Information acquisition and decision making in committees: A survey. *European Journal of Political Economy* 21:563–97.
- Government Accountability Office (GAO). 2016. Proxy advisory firms’ role in voting and corporate governance practices. <https://www.gao.gov/assets/gao-17-47.pdf>
- Greene, W. 2004. The behaviour of the maximum likelihood estimator of limited dependent variable models in the presence of fixed effects. *Econometrics Journal* 7:98–119.

- Hayne, C., and M. Vance. 2019. Information intermediary or de facto standard setter? Field evidence on the indirect and direct influence of proxy advisors. *Journal of Accounting Research* 57:969–1011.
- Iliev, P., and M. Lowry. 2015. Are mutual funds active voters? *Review of Financial Studies* 28:446–85.
- Ingram P., and B. Silverman. 2002. The new institutionalism in strategic management. In *Advances in strategic management*, eds. P. Ingram and B. S. Silverman, 1–30. New York: JAI Press.
- Institutional Shareholder Services (ISS). 2011. 2011–2012 policy survey summary of results. Report, Rockville, MD. <https://www.issgovernance.com/file/files/PolicySurveyResults2011.pdf>
- Jochem, T., G. Ormazabal, and A. Rajamani. 2021. Why have CEO pay levels become less diverse? Working Paper, University of Amsterdam.
- Kakhbod, A., U. Loginova, A. Malenko, and N. Malenko. 2023. Advising the management: A theory of shareholder engagement. *Review of Financial Studies* 36:1319–63.
- Kidd, C. 2023. Can shareholders really trust proxy statements? A negligence standard for Section 14 (a) of the Securities Exchange Act of 1934. *Baylor Law Review* 75:544–63.
- Knyazeva, A., D. Knyazeva, and R. Masulis. 2013. The supply of corporate directors and board independence. *Review of Financial Studies* 26:1561–605.
- Larcker, D., A. McCall, and G. Ormazabal. 2013. Proxy advisory firms and stock option repricing. *Journal of Accounting and Economics* 56:149–69.
- . 2015. Outsourcing shareholder voting to proxy advisory firms. *Journal of Law and Economics* 58:173–204.
- Levit, D., N. Malenko, and E. G. Maug. 2024. Trading and shareholder democracy. *Journal of Finance* 79:257–304.
- Levit, D., and A. Tsoy. 2022. A theory of one-size-fits-all recommendations. *American Economic Journal: Microeconomics* 14:318–47.
- Li, T. 2018. Outsourcing corporate governance: Conflicts of interest within the proxy advisory industry. *Management Science* 64:2951–71.
- Malenko, N. 2014. Communication and decision-making in corporate boards. *Review of Financial Studies* 27:1486–532.
- Malenko, A., and N. Malenko. 2019. Proxy advisory firms: The economics of selling information to voters. *Journal of Finance* 74:2441–90.
- Malenko, A., N. Malenko, and C. Spatt. 2024. Creating controversy in proxy voting advice. Working Paper, Boston College.
- Malenko, N., and Y. Shen. 2016. The role of proxy advisory firms: Evidence from a regression-discontinuity design. *Review of Financial Studies* 29:3394–427.
- McCahery, J., Z. Sautner, and L. Starks. 2016. Behind the scenes: The corporate governance preferences of institutional investors. *Journal of Finance* 71:2905–32.
- McCrary, J. 2008. Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics* 142:698–714.
- Mehta, T. 2013. Shareholder engagement: Maximizing the shareholder relationship. ISS Report, Vol 13.3. https://www.issgovernance.com/file/publications/MaximizingTheShareholderRelationshipVol_13.3.pdf
- Pawliczek, A., A. Skinner, and L. Wellman. 2021. A new take on voice: the influence of BlackRock's 'Dear CEO' letters. *Review of Accounting Studies* 26:1088–136.
- Rouen, E. 2020. Rethinking measurement of pay disparity and its relation to firm performance. *Accounting Review* 95:343–78.

- SEC. 2018. Amendments to the smaller reporting company definition. <https://www.sec.gov/corpfin/amendments-smaller-reporting-company-definition>
- . 2020. Exemptions from the proxy rules for proxy voting advice. Release No. 34-87457. <https://www.sec.gov/rules/final/2020/34-89372.pdf>
- Schapiro, M. L. 2010. Remarks at the NACD Annual Corporate Governance Conference. Speech, October 19. <https://www.sec.gov/news/speech/2010/spch101910mls.htm>
- Shu, C. 2024. The proxy advisory industry: Influencing and being influenced. *Journal of Financial Economics* 154:103810.
- Srinivasan S. 2005. Consequences of financial reporting failure for outside directors: evidence from accounting restatements and audit committee members. *Journal of Accounting Research* 43:291–334.
- Terlaak, A. 2007. Order without law? The role of certified management standards in shaping socially desired firm behaviors. *Academy of Management Review* 32:968–85.
- U.S. Congress. 2023. Oversight of the proxy advisory industry. House Financial Services Committee, July 13. <https://financialservices.house.gov/calendar/eventsingle.aspx?EventID=408896>
- Vafeas, N. 2003. Length of board tenure and outside director independence. *Journal of Business Finance & Accounting* 30:1043–64.
- Whited, R., Q. Swanquist, J. Shipman, and J. Moon, Jr. 2022. Out of control: The (over) use of controls in accounting research. *Accounting Review* 97:395–413.